

Building Segregation: The Long-Run Neighborhood Effects of American Public Housing*

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Abstract

This paper studies the long-term neighborhood effects of the construction of mid-century American public housing. I construct a new national dataset containing the locations, completion dates, and characteristics of the near-universe of public housing projects built between 1935 and 1973, which I link to neighborhood-level data from 1930 to 2010. I document that public housing projects were systematically targeted toward poorer neighborhoods with a higher share of Black residents, consistent with the program's slum clearance goals and racialized site selection politics. Using matched difference-in-differences designs, I find that public housing construction caused large, persistent increases in Black population shares and substantial declines in median incomes and rents in recipient neighborhoods. Public housing construction also appears to have caused racial transition in surrounding neighborhoods, though these estimates are sensitive to the comparison group. Recipient neighborhoods also exhibited lower rates of upward mobility and higher incarceration rates for low-income children who grew up in these neighborhoods. These findings demonstrate that mid-century public housing, despite intentions of neighborhood revitalization, reinforced existing patterns of racial and economic segregation.

JEL Codes: H42, J15, J62, R23, R38

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1 Introduction

The Congress hereby declares that the general welfare and security of the Nation and the health and living standards of its people require housing production and community development sufficient to remedy the housing shortage, eliminate substandard housing through the clearance of slums and blighted areas, and achieve as soon as feasible the goal of a decent home and suitable living environment for every American family.

— *Housing Act of 1949*

Between the 1930s and 1970s, the United States financed the construction of approximately one million federally funded public housing units in one of the most ambitious urban policy initiatives of the 20th century (Shester 2013). Designed to clear slums, address urban housing shortages, and provide affordable housing to low-income families, the American public housing program has become widely synonymous with policy failure. Scholarly and popular accounts have argued that the program created and entrenched racial and economic segregation (e.g., Rothstein 2017; Hirsch 1998; Massey and Denton 1993) and accelerated mid-century urban decline (Jackson 1985). Large megablock projects like Cabrini-Green, Pruitt-Igoe, and the Robert Taylor Homes garnered notoriety and were criticized for concentrating poverty, promoting crime, and destroying the urban fabric of neighborhoods (Jacobs 1961; Newman 1972). Federal policy since the 1970s has reflected this negative view, as the federal government has shifted funding toward market-based alternatives like Housing Choice Vouchers and the Low-Income Housing Tax Credit, while simultaneously dismantling the existing stock through the HOPE VI program, which has funded the demolition or removal of roughly 30% of the original federal public housing stock since the early 1990s (Schwartz 2021).

Yet the evidence for public housing’s failure is mixed. A body of scholarship contends that the program has been unfairly maligned by its most notorious examples and argues that its poor reputation partly reflects broader social and economic challenges faced by American cities in the postwar period (Bauman 1994; Bloom, Umbach, and Vale 2015; Goetz 2013). Indeed, despite widespread criticism of the program, approximately two-thirds of public housing residents reported being “satisfied” or “very satisfied” with their housing as late as 1999 (Schwartz 2021). Some causal evidence from economists suggests that living in public housing has either neutral or positive average effects on individual outcomes (Chaudhry and Eng 2024; Currie and Yelowitz 2000; Jacob 2004; Pollakowski et al. 2022), suggesting that the most negative images of American public housing may not be representative.

At the same time, disentangling the local effects of public housing from pre-existing neighborhood conditions remains challenging. Given that projects were often sited in already-distressed areas as part of slum clearance efforts, did public housing cause neighborhood decline, or simply reflect the targeting of struggling areas? These debates have practical stakes as cities grapple with housing affordability and revisit the question of what role public housing should play.

To shed light on these debates, I construct a new dataset containing the locations, construction dates, and characteristics for the near-universe of public housing projects built from the program's inception until 1973 by combining previously digitized data, administrative data sources, and newly digitized and collected sources. I combine these data with a newly constructed long-run panel of neighborhood characteristics from 1930 to 2010.

I first examine which neighborhood characteristics predicted public housing site selection. Did public housing specifically target Black neighborhoods, as some historical accounts suggest (e.g., [Hirsch 1998](#); [R. Rothstein 2017](#)), or was it more broadly targeted toward poor and working-class neighborhoods? This question matters both for the political economy of public housing and for thinking about how to estimate the effects of public housing construction on neighborhoods. I estimate linear probability models to explore how pre-existing neighborhood characteristics, including demographic, socioeconomic, housing, and institutional factors, predict the placement of public housing projects. I find that public housing projects were more likely to be built in neighborhoods that were initially poorer and more populous, with lower homeownership rates, and a higher share of children, consistent with the program's slum clearance goals, but that even conditional on socioeconomic and housing characteristics, public housing was more likely to be sited in neighborhoods with higher Black population shares, reflecting the racialized politics of housing policy in the mid-20th century. I also find that public housing was more likely to be built in neighborhoods that were later targeted by the urban renewal program, another major mid-century urban policy that targeted predominantly Black neighborhoods ([LaVoice 2024](#)).

I then move to estimate the short- and long-run effects of public housing construction on both the recipient and surrounding neighborhoods. Building on the results from the site selection models, I employ matched difference-in-differences designs that compare each recipient neighborhood to neighborhoods that looked similar before public housing was constructed. Following the approach used by Chetty, Diamond, et al. ([2026](#)) to study the effects of HOPE VI demolitions, I first use nearest-neighbor propensity score matching to select three potential control neighborhoods for each recipient neighborhood based on initial neighborhood characteristics. Then, within each treatment cohort, I pool the se-

lected controls and reweight them to more closely resemble the recipient neighborhoods. I then stack the cohort panels and estimate event studies, comparing changes in recipient neighborhoods with changes in their weighted controls. Identification stems from the fact that federal funding for public housing was limited, and local housing authorities could not build projects in every eligible neighborhood, creating variation in project placement across otherwise similar areas. This stacked framework also avoids the econometric pitfalls that arise in staggered adoption settings when treatment effects are heterogeneous across cohorts and over time (Wing, Freedman, and Hollingsworth 2024). I extend this methodology to analyze the geographic spillovers of the projects by conducting a similar exercise for neighborhoods near recipient neighborhoods, thereby estimating the broader geographic effects of the projects.

In the public housing (“recipient”) neighborhoods, I find sizable increases in Black population shares, driven mainly by growth in the Black population: three decades after construction, the Black population share is 10.3 percentage points higher than in matched control neighborhoods. I also find sizable decreases in median rents and incomes, with estimated changes of -14.6% and -14.9% three decades after construction, along with declines in homeownership rates. These results, in conjunction with the site-selection results, suggest that while public housing was targeted toward poorer neighborhoods with a higher share of Black residents, the construction of the projects further accelerated racial transition and socioeconomic decline in these neighborhoods over the long run.

In nearby neighborhoods, I also find evidence of racial transition, with little evidence of long-run effects on incomes or rents. Using directly matched controls, the Black population share rises by 4.7 percentage points three decades after construction. Here, the shift is driven by declines in the white population, rather than increases in the Black population. Using neighborhoods around the recipient tracts’ matched controls as placebo sites yields smaller Black-share estimates that are not statistically distinguishable from zero, so I interpret these nearby effects more cautiously.

I also explore the heterogeneity of these effects along several dimensions, beginning with initial neighborhood demographics. First, I ask whether neighborhood effects depend on the initial share of Black residents in the neighborhood pre-construction, to test whether there is evidence that public housing led to racial “tipping” dynamics, as some historical accounts have suggested (R. Rothstein 2017). To test this, I group neighborhoods into three groups: nearly all-white neighborhoods (below 1% Black share), neighborhoods in what I call a plausible “tipping range” (between 1% and 12%, based on the tipping-point estimates of Card, Mas, and J. Rothstein (2008) for the 1970s), and above 12%. I find little evidence of tipping in the sense of a distinct response in the intermediate range. Recipi-

ent neighborhoods that began with Black population shares above 1% lost white residents and experienced larger declines in income whether their initial share was in the tipping range or well above it, and Black population shares rose in all three groups. The sharpest contrast is at the bottom: nearly all-white recipient neighborhoods did not lose white residents, their populations grew, and their incomes fell less. Nearby neighborhoods in the 1% to 12% range lost the most white residents in absolute terms, but initially Black neighborhoods had fewer white residents to lose, and in proportional terms the two groups are not distinguishable. Second, I ask whether the effects depend on a neighborhood's initial income, comparing recipient neighborhoods whose pre-construction median income fell in the bottom tercile of the matched panel with the rest. I find that the effects of public housing construction are concentrated in the poorest neighborhoods, which experienced larger increases in Black population share, lost a substantial share of their white residents, and saw their median incomes and rents fall, whereas higher-income recipient neighborhoods gained Black residents without losing white residents and saw much smaller declines in income. Nearby neighborhoods show the same pattern.

I then turn to estimate differences by characteristics of the projects themselves. I find that larger projects do not have larger effects on racial composition, but that they do drive larger socioeconomic changes. I also find that the long-run effects are larger for projects built in the 1940s and 1950s than for later projects, consistent with shifts towards smaller projects in the 1960s and onward.

Finally, I explore the potential implications of these neighborhood effects for long-run economic opportunity. To do so, I link my dataset to Opportunity Atlas data on the upward mobility and later incarceration rates of children from low-income families born between 1978 and 1983 ([Chetty, Friedman, et al. 2018](#)). I find that recipient tracts have lower upward mobility and higher incarceration rates than their weighted controls. After conditioning on 1980 neighborhood characteristics, neither the recipient mobility differences nor the incarceration gaps are statistically distinguishable from zero, suggesting that the relationship between public housing and these childhood outcome measures is largely explained by public housing-driven changes in neighborhood composition. I find limited evidence of worse outcomes in nearby neighborhoods.

This paper relates to several strands of the literature in urban and public economics and economic history. First, it contributes to work on the local effects of affordable housing programs ([Baum-Snow and Marion 2009](#); [Diamond and McQuade 2019](#)) and neighborhood revitalization policies ([Collins and Shester 2013](#); [LaVoice 2024](#); [Rossi-Hansberg, Sarte, and Owens 2010](#); [Blanco and Neri 2025](#)) by estimating the neighborhood effects of one of the largest urban policies in American history. More directly, studies of HOPE

VI-era public housing demolitions find that removing distressed projects reduced nearby crime, increased housing demand, changed neighborhood racial and income composition, and generated uneven welfare effects across groups (Tach and Emory 2017; Blanco 2023; Aliprantis and Hartley 2015; Sandler 2017; Almagro, Chyn, and Stuart 2023). Related work following displaced children finds long-run gains in employment and earnings, especially for children moved out of the most distressed projects (Haltiwanger et al. 2024; Chyn 2018), and also finds that HOPE VI may have created higher opportunity neighborhoods (Chetty, Diamond, et al. 2026). This literature has focused on the demolitions of the most distressed public housing projects, and much of it has focused on a single city (Chicago). There are several reasons to think these estimates of the effects of demolitions may not generalize to those of public housing construction. First, the projects demolished through the HOPE VI program are a selected set of the most distressed projects. They are thus not necessarily representative of the public housing program as a whole. Second, the effects of project construction likely depended on existing neighborhood conditions, which likely varied in ways that they did not for demolitions. New projects may have served as local amenities in some neighborhoods but disamenities in others. Finally, the effects observed in Chicago may not generalize to other cities.

My findings also complement earlier work in other social sciences on public housing and the concentration of poverty in central cities, which generally used data compiled from one or a few cities (e.g., Carter, Schill, and Wachter 1998; Massey and Kanaiaupuni 1993). Relative to this older literature, I use a dataset covering a much broader set of cities, projects, and time periods, and apply modern econometric techniques to estimate the causal effects of public housing construction on neighborhoods.

Most directly, this paper contributes to the literature estimating the effects of public housing construction. To study the effects of public housing construction on a variety of county- and city-level outcomes, Shester (2013) uses digitized HUD data covering all federally funded public housing projects built through 1973, with locations only available at the locality level (discussed in Section 4). That paper finds that cities in the same state that experienced more public housing construction saw decreases in median property values, median family income, and population density. Shester, Allen, and Handy (2019) uses the same dataset to study the effects of public housing construction on the rise in single motherhood at the MSA level. I build on this work by merging precise locations with these data, allowing me to study the neighborhood effects of public housing construction.

My paper fits into an emerging literature on the neighborhood-level effects of public housing construction. Günnewig-Mönert (2025) studies the effects of public housing construction in New York City from 1930 to 2010 on neighborhood racial composition, rents,

and welfare, and how these effects differed based on the design of the projects. My paper builds on this work by assembling project-level data for the entire country to characterize the public housing program more broadly. This scope matters because the New York City Housing Authority may have been an outlier among American housing authorities, having largely avoided the management failures and high-rise demolitions that defined the program elsewhere (Bloom 2008; Schwartz 2021). Estimates from New York City alone, therefore, may not be representative of the program as a whole. Two other recent papers study the initial effects of public housing construction during the first decades of the program. Harris (2025) studies the local effects of public housing constructed from 1940 to 1960 and, using a shift-share IV approach, finds that public housing initially reduced the concentration of poverty in recipient neighborhoods. Blanco and Perdoni (2026) study the initial public housing constructed during the New Deal and find limited effects on broader neighborhood composition. My paper both studies a broader set of public housing projects and documents the long-run effects as the public housing program evolved over time.

Finally, my paper contributes to the economic literature on the emergence and consequences of segregation in the United States. While prior research has documented migration responses to neighborhood demographic change (Shertzer and Walsh 2019; Boustan 2010), my paper addresses one way in which federal housing policy directly influenced neighborhood sorting. More broadly, this paper contributes to long-standing debates around the role that public policy has played in shaping racial and economic segregation in American cities (R. Rothstein 2017; Trounstein 2018; Boustan 2013; T. D. Logan and Parman 2025).

The paper proceeds as follows. Section 2 describes the history of the public housing program. Section 3 outlines the potential channels through which public housing may affect neighborhoods. Section 4 describes data sources and construction. Section 5 analyzes the determinants of public housing site selection. Section 6 presents the empirical strategy and estimates of neighborhood effects. Section 7 explores heterogeneity in these effects and examines potential mechanisms. Section 8 links public housing to long-run economic opportunity using data from the Opportunity Atlas. Section 9 concludes.

2 Historical Background: The U.S. Public Housing Program

Public housing in the United States began during the Great Depression as part of New Deal efforts to reduce urban housing shortages and stimulate the economy. The Public Works Administration built the first projects, producing about 20,000 units before court

challenges over land acquisition limited its scope ([Jackson 1985](#)). The program was greatly expanded and decentralized with the passage of the Housing Act of 1937 (the “Wagner-Steagall Act”), which promoted the creation of local Public Housing Authorities (PHAs) to build and oversee public housing with federal funds. Specifically, PHAs were responsible for choosing public housing locations, building the projects, and managing them. At first, federal funding was limited to funding the construction of the projects, while operating costs were meant to be covered by tenant rents ([McCarty 2014](#)). The 1937 Act’s stated purpose was slum clearance and redevelopment rather than expanding housing supply, which reflected contemporary beliefs that “slum” neighborhoods reduced nearby property values and caused poor health and behavior among residents ([Meyerson and Banfield 1955](#)). In this vein, every newly constructed unit required the elimination of an unsafe or unsanitary dwelling ([McCarty 2014](#)).

The program was greatly expanded as part of a broad urban policy agenda following World War II. Many Northern cities experienced severe housing shortages, which were especially acute for African Americans, while white flight, suburbanization, and urban “blight” challenged cities. These challenges generated a large political coalition for urban redevelopment policy, culminating in the Housing Act of 1949 ([von Hoffman 2000](#)). Title III of the 1949 Act provided funding for an additional 810,000 units of public housing and marked a major shift in the program’s scope and goals, expanding public housing as a tool for slum clearance, a source of low-income housing, and a potential destination for those displaced by the urban renewal slum clearance program, which was funded by Title I of the Act, since localities were now required to give displaced families preference for public housing units ([McCarty 2014](#)).

The ambitious goals of the 1949 Act, however, quickly met political, fiscal, and social obstacles, and several developments in the 1950s and 1960s transformed public housing into a deeply controversial policy. First, as detailed in Section 2.1, racial dynamics in site selection led to intense political conflicts. Second, the composition of public housing tenants changed significantly over time. Early projects primarily housed working families and the “deserving poor,” but by the 1960s, they increasingly housed the poorest and most disadvantaged households. The rise of suburban homeownership allowed working-class families to leave public housing, while authorities tightened income eligibility limits. Federal regulations also mandated that PHAs prioritize the neediest applicants, resulting in the concentration of poverty within projects ([Schwartz 2021](#)). Third, design and construction issues became more evident. To avoid competing with private housing and to meet mandated construction cost limits, projects were intentionally made austere and inexpensive, making them susceptible to rapid deterioration ([Schwartz 2021](#)). Many pub-

lic housing authorities struggled to fund maintenance as middle-class tenants exited the projects and rent revenues declined. Furthermore, the 1969 Brooke Amendment capped tenant rents at 25% of income, further limiting public housing authorities' ability to fund necessary maintenance with revenues from tenant rents (Hunt 2018). These challenges are illustrated by well-known public housing failures, such as St. Louis's Pruitt-Igoe Homes, completed in 1956. The modernist mega-project was initially hailed as a model of public housing success, but within several years, it began to fall into disarray, beset by crime, vandalism, and abandonment as maintenance funding dried up and higher-income residents moved out (Bristol 1991). In response to these issues, the complex was ultimately demolished beginning in 1972.

In response to mounting criticism, American housing policy has shifted away from direct public housing provision. A 1971 report by the Nixon Administration stated that "drab, monolithic housing projects, largely segregated...still stand in our cities as prisons of the poor" (Orlebeke 2000). In 1973, President Nixon declared a moratorium on subsidies for traditional public housing. By the early 1990s, the public housing stock faced serious challenges: physical deterioration due to deferred maintenance, extreme concentration of poverty as working families moved out, and, in some cases, rampant crime and gang activity. In response to these issues, the federal government began the HOPE VI program in 1992, which provided federal grants to demolish or redevelop distressed public housing projects and transform them into mixed-income developments. Meanwhile, the Faircloth Amendment of 1998 has capped the total number of units that public housing authorities can operate at their 1999 levels, further limiting the growth of public housing.

2.1 Race, Segregation, and Public Housing

Race and segregation emerged as central and contentious issues throughout the program's history. From the outset, the Public Works Administration followed a so-called "neighborhood composition rule," formally segregating projects based on the demographics of the neighborhoods where they were built. While ostensibly a neutral policy intended to maintain neighborhood "stability," in practice, the rule may have solidified segregation. Notable accounts, such as R. Rothstein (2017), contend that public housing not only reinforced existing segregation but also actively created it by constructing segregated projects in previously integrated neighborhoods. A well-known example is Atlanta's Techwood Homes, built in 1936 as an all-white development, which displaced a previously integrated low-

income neighborhood (R. Rothstein 2017).¹ Many large public housing authorities continued to follow an explicit neighborhood composition rule until the 1960s.² However, even after these legal changes, projects in many cities remained highly segregated in practice (Bickford and Massey 1991).

Historical case studies show that issues related to race and segregation also influenced site selection decisions, and contemporaneous debates among policymakers reveal that they were fully aware of the program’s potential to either entrench or reduce existing residential segregation patterns (Hirsch 2000). In Chicago, the housing authorities had initially planned to build projects throughout the city, but strong opposition from white neighborhoods led to the concentration of projects in predominantly Black areas on the South and West Sides (Meyerson and Banfield 1955; Hirsch 1998). Similar conflicts occurred in other major cities, with white residents and politicians successfully blocking projects in their neighborhoods. I examine a specific example of this dynamic in Philadelphia, comparing proposed-but-not-selected areas to public housing locations in Section 5.2. Building on this historical evidence, Section 5.1 tests whether these racial site selection dynamics held systematically nationwide.

2.2 Motivating Example: Hunters Point in San Francisco

The history of Bayview–Hunters Point in San Francisco provides a clear example of how public housing construction can change a neighborhood’s long-run trajectory. Before World War II, the area was a racially mixed neighborhood on the city’s periphery. During the war, the federal government built thousands of temporary housing units there to accommodate defense workers, which led to an initial influx of African American families into the neighborhood. After the war, rather than dismantling these temporary units, local and federal officials converted many of them into permanent public housing. Throughout the 1950s and 1960s, the San Francisco Housing Authority replaced and expanded these units into large-scale projects and moved Black families into these developments. This expansion coincided with a rapid demographic shift: as the projects grew, the neighborhood ceased to be a mixed-income area; white and Asian households left, and the community

¹By contrast, recent work by Blanco and Perdoni (2026) finds limited evidence that these early public housing projects generally had an effect on broader neighborhood racial composition.

²Legal challenges against segregation in public housing began in the mid-1950s, including the U.S. Supreme Court’s 1954 decision to allow California courts to ban segregation in San Francisco’s housing projects. However, federal law did not formally prohibit race-based segregation in public housing until the 1960s. An executive order in 1962 directed federal agencies to prevent racial discrimination in federally-owned housing, and Title VI of the Civil Rights Act of 1964 prohibited discrimination in any program receiving federal financial assistance (McCarty 2014).

became increasingly Black and segregated. Figure 1 shows the racial transition of these neighborhoods alongside the locations of public housing projects, showing a sharp rise in the Black population share and the decline in the white population from 1940 to 1970. The rest of this paper tests whether the patterns seen in Hunters Point were systematic across the country.

3 Potential Neighborhood Effects of Public Housing

To fix ideas, I briefly outline the main channels through which the construction of public housing projects might have affected neighborhood outcomes in the short and long run. First, public housing construction may have mechanically altered the racial and socioeconomic composition of neighborhoods, because tenant selection policies, income limits, and subsidized rents determined who occupied these projects and thus the demographic makeup of the areas where they were built. Second, these compositional changes may have triggered endogenous sorting by private households: if individuals have preferences for their neighbors' race or income, the arrival of public housing residents might have prompted some existing residents to relocate. Prominent historical accounts such as Jackson (1985) argue that public housing drove "white flight" from central cities, though empirical evidence on this point remains limited.

Third, the physical structure of public housing projects could create built-environment externalities: replacing substandard or vacant structures with new housing could improve neighborhood quality and desirability, as shown for private housing upgrades by Rossi-Hansberg, Sarte, and Owens (2010) and for subsidized housing by Ellen et al. (2007). Conversely, large superblocks or architecturally incongruous high-rise towers might be viewed as local disamenities. Indeed, criticism of the mid-century program has often focused on the design of the projects (Jacobs 1961; Newman 1997). Fourth, public housing construction may generate broader social and market externalities. A frequent concern is that concentrated poverty within large projects generates social disorder and elevated crime, with potential spillovers into surrounding neighborhoods (Sandler 2017). The direction of these effects may vary across contexts: modern evidence on the LIHTC program suggests that subsidized housing investments can be a positive amenity in distressed areas but a disamenity in more affluent ones (Diamond and McQuade 2019).

4 Data

4.1 Public Housing Data

A significant challenge in studying the history of public housing is the lack of a comprehensive public dataset that includes project lists, construction dates, and precise project locations. Consequently, previous research on the neighborhood effects of public housing has largely been limited in scope or restricted to a small number of cities where researchers obtained this information directly from housing authorities. The data limitations have also hindered broader historical analyses of the public housing program, and have been acknowledged by previous work ([Ellen et al. 2007](#); [Hunt 2018](#)). The administrative data that do exist were largely compiled in recent decades, so projects that were demolished or otherwise ceased to exist, disproportionately among the most distressed projects, are likely to be missing from these datasets. My paper addresses this gap by constructing what I believe is the most complete dataset of mid-century public housing by combining information from a series of administrative datasets with previously digitized and newly digitized sources.

The first source I use is the *Consolidated Development Directory* (CDD), published by the U.S. Department of Housing and Urban Development (HUD) in 1973 and digitized by [Shester \(2013\)](#). The CDD lists all federally funded public housing projects that existed in 1973 and includes project characteristics, including unit counts and, crucially, project completion dates.³ The CDD does not contain location information for the projects, so previous work using these data ([Shester 2013](#); [Shester, Allen, and Handy 2019](#)) has been limited to studying aggregate city- and county-level outcomes. These data also contain federal project codes, which I use to link these records to later HUD datasets that contain building addresses and project coordinates.

I supplement the CDD with historical project inventories obtained from libraries, housing authorities, and public records requests. I collected these sources for two reasons. First, the CDD covers only the federal program: little data exist on city- and state-funded public housing, and New York City in particular is known to have had a notable number of projects funded by the city and state. Second, in cities where both sources exist, the inventories provide an independent check on the CDD's project lists and completion dates.

³The directory also records leased housing projects, privately owned housing that local authorities leased and subleased to low-income families under the Section 23 program, which began in 1965. I exclude these leased projects from my analysis. A description of the Section 23 program can be found in National Housing Law Project ([2012](#)). I also exclude housing built under the federal Indian Housing Program, which was administered separately through tribal housing authorities.

I locate projects by compiling a national dataset of individual public housing building addresses from several sources: the building-level address file underlying HUD’s *Subsidized Housing Projects’ Geographic Codes* (Form HUD-951), which records the addresses of roughly 288,000 public housing buildings surveyed in 1988–89 ([U.S. Department of Housing and Urban Development 1996](#)); HUD’s National Geospatial Data Asset (NGDA) building inventory in 2023, from which roughly 111,000 currently existing buildings enter my address dataset⁴; the 2016 New York City Housing Authority *Address Guide*; and a list of the addresses of all public housing projects that were owned by the Chicago Housing Authority, provided to me by D. Bradford Hunt. I also digitized the building addresses for the Pruitt-Igoe project in St. Louis from its 1964 site plan. I geocode these addresses using ESRI’s StreetMap Premium geocoder, and link buildings to the CDD projects using the federal project codes described above, along with name-based matching in New York City. Since some mid-century projects were quite large, using building-level addresses lets me capture the full project footprint, rather than just a single project address. For projects for which I cannot locate a usable building address, I use project-level coordinates from HUD’s Picture of Subsidized Households (PSH) datasets from 1997 and 2000, which report coordinates for federally funded housing projects from 1997 onward.

Overall, I identify the locations of almost the entire universe of mid-century public housing in the United States. I geolocate 99% of the total completed public housing units in the CDD, and the projects that I am unable to locate are predominantly small, with a median of 25 units. In my ultimate analysis sample, 81% of projects are located through actual building addresses, and the remainder through project-level coordinates.

I also supplement the project inventory with information on project population and racial composition. For most cities, these measures come from the 1977 Picture of Subsidized Households dataset, which was cleaned and shared with me by Yana Kucheva ([Kucheva 2013](#)).⁵ For New York City, I use population-by-race data from the early 1970s digitized and shared with me by Max Günnewig-Mönert ([Günnewig-Mönert 2025](#)), and for Chicago, I obtained population-by-race data from the 1973 digitized Annual Report of the Housing Authority. While it would be ideal to measure population by race at the time of construction, these data are not available in most cities. I therefore use the 1970s values as a proxy for the population and racial composition of each project at the time of construc-

⁴HUD revamped its project codes in 2008, meaning that a direct match between NGDA project codes and CDD project codes may mismatch projects. I construct a crosswalk between the 2007 and 2008 project codes to do this mapping, described in Appendix A.

⁵These data report the number of subsidized households by race in each project, but do not include direct population counts. In Appendix A, I describe how I convert these household counts to population counts.

tion. I ultimately use these data to estimate each neighborhood’s private population in each decade by subtracting the estimated public housing population from its Census population. As a retrospective measure of project distress, I also link projects to HUD’s roster of HOPE VI revitalization grants awarded between 1993 and 2010, obtained from the site list published by Chetty, Diamond, et al. (2026).⁶ I match award development names to my projects within each state and city and resolve ambiguous cases through manual review. Eventual HOPE VI selection serves as a retrospective marker of projects that later became especially distressed and received federal funding for demolition or redevelopment.

4.2 Neighborhood Data

To study the neighborhood effects of public housing construction, I construct a panel dataset of neighborhood characteristics from 1930 to 2010.

First, I collect a variety of outcomes from the 1930 to 2010 decennial census tract tables, which I download from IPUMS NHGIS (Manson et al. 2022). These data include tract-level population counts by race (white, Black, and other), age structure (share under 18 and share 65 and older), several socioeconomic measures (median income, high school graduation rates, labor force participation, and unemployment rates), and housing market measures (median contract rents, homeownership rates, and the share of housing units needing major repairs).⁷ I also measure each tract’s distance to the nearest central business district (CBD), using the CBD indicator published in the 1980 census tract tables.

I harmonize all observations to 1950 census tract boundaries using an area-reweighting procedure (Eckert et al. 2020). I choose to harmonize to an early base geography to limit the scope for public housing-induced changes in tract boundaries to affect my analysis (Glaeser, Gyourko, and Neiszner 2025). I choose 1950, rather than an earlier year, to avoid issues with the 1940 New York City NHGIS tract shapefiles, which represent larger Health Districts rather than Census tracts. Appendix B.2 describes the tract harmonization process in more detail.

I group census tracts into core-based statistical areas (CBSAs) using the Office of Management and Budget’s June 2003 delineations, based on Census 2000 population and commuting data (Office of Management and Budget 2003). I assign tracts through their counties and hold these geographic assignments fixed throughout the 1930–2010 panel, providing a consistent reference geography for the metropolitan comparisons.

⁶Available at <https://opportunityinsights.org/paper/hopevi/>

⁷I draw the share of units in need of major repairs from the 1940 tract tables. I use this measure only for 1940, as it is not consistently available in other decades.

I supplement the published tract-level data with full-count Census microdata from 1930 and 1940 ([Ruggles et al. 2022](#)). This allows me to expand geographic coverage beyond cities and neighborhoods with published tract-level data in 1930 and 1940 and to construct measures of neighborhood income, which are not available in the census tract tables for those years. To aggregate the individual-level data to 1950 Census tract level, I first aggregate to the enumeration district level using the 1930 and 1940 enumeration district shapefiles from the Urban Transitions Project ([J. R. Logan et al. 2024](#)). I then use area weighting to concord the enumeration district-level data to 1950 Census tracts. The 1930 Census did not collect income data, so for that year I sum the machine-learning-adjusted occupation scores developed by Saavedra and Twinam (2020) across family members. For 1940, I construct family wage income by summing individual wage and salary earnings. I express all monetary values in 2000-dollar terms using the U.S. CPI from Officer and Williamson (2025).

I supplement the Census panel with three sets of historical geographic covariates. First, I incorporate the digitized data and shapefiles of the Home Owners' Loan Corporation (HOLC) "redlining" maps drawn in the late 1930s, made available by *Mapping Inequality* ([Nelson and Winling 2023](#)). These maps graded neighborhoods from "A" (best) to "D" (hazardous) and capture perceived credit risk and potential discrimination.⁸ Following Weiwu (2025), I classify a neighborhood as "redlined" if at least 80% of the area is designated as "hazardous" (HOLC grade D). Second, I incorporate data on the locations of U.S. urban renewal projects (1955–1966) from *Renewing Inequality* ([Nelson and Ayers 2025](#)). Urban renewal, established under Title I of the 1949 Housing Act, provided federal subsidies for cities to acquire and clear "blighted" neighborhoods. Urban renewal was closely intertwined with public housing: clearance projects often created sites for new developments or displaced families that public housing was intended to rehouse ([von Hoffman 2000](#)).⁹ To account for potentially confounding effects of urban renewal and to explore the relationship between the two programs, I overlay these maps onto 1950 census tract boundaries and classify a tract as an "urban renewal" tract if at least 5% of its area overlaps with an urban renewal project. Finally, I incorporate data on the locations of interstate highways as of 1996 from Weiwu (2025). Interstate highway construction began in the late 1950s

⁸Although many scholars have argued that these maps institutionalized redlining and curtailed investment in minority neighborhoods ([Aaronson, Hartley, and Mazumder 2021](#)), recent evidence complicates this view. Fishback et al. (2022) shows that the maps largely reflected existing patterns of racial and socioeconomic segregation rather than shaping new lending decisions.

⁹Recent research finds that Black neighborhoods were two to three times more likely to receive urban renewal projects than white neighborhoods, and that urban renewal areas experienced declines in housing and population density alongside rising rents and incomes ([LaVoice 2024](#)).

and has been accused of disproportionately displacing minority communities (Rose and Mohl 2012). I calculate the distance from each tract to the nearest interstate to examine the relationship between highway locations and public housing placement, and to assess the potential confounding effects of highway construction on neighborhood change.

Finally, I measure long-run individual outcomes using the Opportunity Atlas (Chetty, Friedman, et al. 2018), which reports adult outcomes by childhood census tract for children born between 1978 and 1983. I use two outcomes, pooling across race and gender: mean adult household income rank, measured in 2014–2015, and the incarceration rate on April 1, 2010. For both outcomes, I use the Atlas’s estimates for children whose parental income was at the 25th percentile of the national income distribution. I refer to these children as “low-income children.” These outcomes are available at the 2010 Census tract level, so I again concord these outcomes from 2010 to 1950 census tract boundaries using my geographic crosswalks and the counts of children from the Opportunity Atlas.

4.3 Defining the treatment

To study the effects of public housing construction on neighborhoods, I define a set of recipient neighborhoods that received public housing projects, as well as a set of nearby neighborhoods that did not receive public housing directly but were most likely to experience geographic spillovers due to their proximity to a public housing project.

First, to ensure that a project represents a meaningful potential intervention at the census tract level, I exclude projects with fewer than 50 housing units. I also exclude “scattered site” projects, which are often listed under a single project identifier but actually represent housing units located throughout a city. I identify scattered sites by name, or, for projects with at least five geocoded buildings, when the median distance of the buildings from their own median location exceeds one kilometer.¹⁰

I define recipient tracts based on the locations of the individual buildings that make up each project. When a project’s buildings are geocoded, I assign each building to the tract it falls within, and consider a tract a recipient tract if it contains at least one building from a qualifying public housing project. For projects where I was only able to geocode a single project-level coordinate, I consider any tract intersecting a 100-meter buffer around it as a recipient tract. This buffer captures the fact that some projects are large enough to straddle multiple tracts. Under either rule, a project may span several tracts. When it does, I allocate its units and population across them in proportion to the number of buildings

¹⁰This threshold was chosen by hand, and does not drop ultra-large contiguous developments in the data, such as the Robert Taylor Homes corridor in Chicago.

in each tract. For projects assigned based on the project-level coordinate, I divide them evenly.

Since the Census data are decennial, I assign treatment timing by decade. Following the literature, I use the project’s completion date as the treatment date ([Asquith, Mast, and Reed 2023](#)). For CDD projects, I use the year of HUD’s “date of full availability,” the month in which substantially all of a project’s units became available for occupancy. For projects from the city inventories, I use the completion year reported in the inventory. I assign a recipient tract to cohort t if its first qualifying project was completed between $t - 9$ and t (e.g., tracts receiving projects between 1951 and 1960 belong to the 1960 cohort). This timing convention ensures that the treatment occurs after the pre-treatment observation but before the post-treatment observation in the panel.¹¹ For tracts that receive multiple public housing projects over time, treatment timing is determined by the first project that meets the size threshold. Subsequent projects in the same tract are not considered separate treatment events, since the tract has already received a qualifying public housing project.

To examine spatial spillovers, I define a nearby tract as one that never received a qualifying public housing project but whose boundary lies within one kilometer of at least one building of a qualifying project. For projects located only by project-level coordinates, I measure the distance to that coordinate instead. Nearby tracts inherit the earliest treatment year among their recipient neighbors, and a tract within one kilometer of several projects is considered nearby the project completed earliest. I adopt a one-kilometer threshold based on prior literature showing the geographic extent of spatial spillovers from public housing redevelopment ([Blanco and Neri 2025](#)). Figure 2 illustrates the recipient and nearby classification for Chicago.

4.4 Sample Construction

My empirical analysis focuses on projects built between 1941 and 1973, with neighborhood outcomes measured from 1930 to 2010. I restrict the sample in several ways to ensure a balanced panel of neighborhoods with sufficient pre-treatment data to conduct my difference-in-differences analysis.

Table 1 reports the sample restrictions and resulting attrition. I require tracts to appear every year from 1930 to 2010 and to have non-missing values for all outcomes and covariates in each decade. I exclude tracts that received public housing before 1941 to preserve 1930 and 1940 as clean pre-treatment periods in my difference-in-differences analysis. I

¹¹If a project was completed in the Census year but after the Census enumeration date, the effects of the project would meaningfully be captured in the second, rather than first, post-treatment decade, and the $t = 0$ effects would be understated.

also drop metropolitan areas with fewer than 30 total tracts to ensure sufficient within-city variation. I also exclude neighborhoods that were initially sparsely populated (populations below 250 in 1930 or 1940) and geographic outliers (tracts exceeding 10 km², which are typically peripheral suburban tracts).

The resulting balanced sample includes 7,427 census tracts across 48 CBSAs, representing 30.8 million people in 1940 (approximately 23% of the U.S. population). This sample contains 976 public housing projects with 387,905 total units located in 1,003 recipient tracts. Figure 3 shows the geographic distribution of CBSAs included in the analysis. These sample restrictions have implications for external validity. Due to data limitations, the analysis focuses on neighborhoods in major metropolitan areas and excludes public housing built in rural areas or smaller cities, as well as neighborhoods that were not populated at the beginning of the period. By excluding projects with fewer than 50 units, my analysis also does not speak to the effects of smaller-scale or scattered-site public housing. Finally, my analysis excludes the earliest public housing projects, including the New Deal projects studied by Blanco and Perdoni (2026). Appendix Table A1 compares included and excluded projects and shows that the analysis sample disproportionately includes larger developments and projects in the Northeast and Midwest.

5 Site Selection

In this section, I study the placement of the public housing projects in my sample. These dynamics matter for several reasons. First, the site-selection process for mid-century public housing was a significant criticism of the program. Historical case studies in several cities suggest that projects were often targeted to poorer and minority neighborhoods, both because of the program's slum clearance goals and because of local backlash in white neighborhoods against construction of projects (Meyerson and Banfield 1955; Bauman 1994; Sugrue 2005). Second, these site selection dynamics have had important legal implications: Lawsuits in several cities have alleged that public housing site selection was discriminatory and in violation of civil rights law, most famously *Gautreaux v. Chicago Housing Authority* (1966), but also in cities like Dallas (*Walker v. HUD*, 1985) and Baltimore (*Thompson v. HUD*, 2005). Still, there has been little systematic evidence on the nationwide patterns of public housing site selection. Finally, understanding these site-selection dynamics is important for estimating the neighborhood effects of public housing and interpreting those effects.

To begin, in Section 5.1, I estimate which pre-existing neighborhood characteristics predict the eventual locations of public housing projects. I also test whether the placement

of public housing projects was related to the locations of other mid-century urban policies, particularly urban renewal and the interstate highway system. Then, in Section 5.2, I zoom in on a particular case study of site selection by examining the locations of proposed-but-not-built public housing in Philadelphia. This example illustrates the political dynamics around site selection decisions.

5.1 Where were the projects built?

Table 2 shows the raw comparison of 1940 baseline characteristics between census tracts that eventually received a qualifying public housing project and those that did not. Recipient neighborhoods had substantially higher Black population shares, lower median incomes and rents, higher unemployment rates, a higher share of homes needing major repairs, and were located closer to a central business district (CBD).

To formally test which neighborhood characteristics predict public housing placement, I estimate linear probability models relating pre-treatment tract characteristics to the likelihood of ever receiving a public housing project. Specifically, I estimate:

$$\text{Recipient}_{im} = \gamma_m + \mathbf{X}'_{i,1940}\boldsymbol{\beta} + \mathbf{Z}'_i\boldsymbol{\delta} + \epsilon_{im} \quad (1)$$

where Recipient_{im} is an indicator for whether tract i in CBSA m received a qualifying public housing project between 1941 and 1973, using the project eligibility and tract assignment rules described in Section 4.3. The vector $\mathbf{X}_{i,1940}$ contains 1940 neighborhood characteristics, and $\mathbf{Z}_i$ includes distance to the nearest interstate highway and an indicator for whether a tract was treated by urban renewal. CBSA fixed effects γ_m absorb differences in average placement rates across metropolitan areas, so the estimates compare neighborhoods within the same CBSA.

Columns (1)–(3) of Table 3 estimate the model without the $\mathbf{Z}_i$ controls, while Column (4) adds them to examine whether the relationship between 1940 characteristics and public housing placement is robust to controlling for subsequent federal interventions. I adjust standard errors for spatial correlation following Conley (1999), allowing for correlation of residuals across census tracts within a 2-kilometer radius.

I chose variables based on historical narratives of public housing site selection and the program’s stated intentions. First, as described in Section 2, the public housing program was largely intended as a “slum clearance” program, so we might expect that neighborhoods with lower socioeconomic status and worse housing market conditions would be more likely to receive public housing. Second, historical case studies in multiple cities have documented the key role of race in determining where public housing was built (e.g.,

[Hirsch 1998](#); [Bauman 1994](#)), so I include the Black population share as a key predictor of public housing placement. I also include the HOLC redlining designation to capture long-term patterns of racial segregation, discrimination, and disinvestment.

The results from these linear probability models are shown in Columns (1)–(4) in Table 3. Column (1) shows a parsimonious model with several key neighborhood characteristics, while Column (2) shows a more saturated model including all neighborhood characteristics, including homeownership rates and age composition. Column (3) adds the share of housing units deemed in need of major repairs in 1940, which is missing for about 5% of tracts. Consistent with the historical narrative, I find that public housing projects tended to be targeted toward poorer, minority neighborhoods: census tracts that were initially more populous, had higher Black population shares, lower median incomes and labor force participation rates, and had higher unemployment rates were more likely to receive public housing projects during this period. Neighborhoods with lower homeownership rates and a higher share of children were also more likely to receive public housing. Redlining is also positively associated with public housing receipt, although the estimates are statistically significant only at the 10% level. The housing disrepair coefficient in Column (3) and the rent coefficient in Column (4) are not statistically distinguishable from zero at the 5% level. These findings reflect the slum clearance motivation of the public housing program and are also consistent with historical accounts emphasizing the racial targeting of public housing site selection: Even after controlling for local economic, housing market, and demographic conditions, the Black population share remains a strong and significant predictor of public housing placement.

Column (4) additionally tests whether neighborhoods that ultimately received public housing were more likely to be affected by two other transformative mid-century urban policies: urban renewal and the interstate highway system. Evidence and historical narrative suggest that these programs may have affected many of the neighborhoods that were also targeted for public housing. The urban renewal program, in particular, was directly related to public housing in many cities, as public housing projects were used to house individuals displaced by urban renewal ([Bauman 1994](#); [Hirsch 1998](#)). Neighborhoods that received public housing were also much more likely to be affected by urban renewal, whereas proximity to an interstate highway was not significantly related to public housing placement. This result further confirms the interplay between these two programs and motivates accounting for urban renewal in my empirical strategy in Section 7.4.¹²

¹²These results are consistent with similar exercises in Harris (2025) and Massey and Kanaiaupuni (1993).

Quantitatively, these effects are sizable. The baseline probability that a census tract in the sample received a public housing project between 1941 and 1973 is 13.5%. Based on the estimates in Column (4), a one standard deviation increase in Black population share is associated with a 4.1 percentage point (30.7%) higher public housing selection probability, a one standard deviation increase in unemployment rate with a 1.8 percentage point (13.3%) higher probability, and a one standard deviation decrease in log median income with a 4.9 percentage point (36.3%) higher probability. Finally, neighborhoods that were eventually targeted by urban renewal were 7 percentage points (52%) more likely to receive public housing.

5.2 A Philadelphia Site Selection Case Study

Historical accounts of public housing site selection have highlighted the political battles that surrounded the placement of projects in particular cities, for example, in Chicago (Meyerson and Banfield 1955), Philadelphia (Bauman 1994), and Detroit (Sugrue 2005). These accounts have emphasized the conflict between public housing authorities and white working-class neighborhoods that resisted the construction of public housing projects in their communities. Here, I present a case study of site selection in Philadelphia, drawing on the historical treatment and maps from Bauman (1987). This case study is illustrative of the political dynamics that shaped site selection decisions in many cities (Hunt 2018).

Following the 1954 Housing Act, the Philadelphia Housing Authority proposed 21 public housing projects across the city. Upon announcement, many of these proposed projects faced significant backlash from local white communities, leading to their eventual cancellation. I identified these proposed Philadelphia public housing locations by scanning and georeferencing a historical map from Bauman (1987), shown in Figure 4. I georeferenced the historical map in QGIS, aligning it with a modern basemap using identifiable landmarks, such as major roads and rivers. Then, I located the proposed public housing sites on the georeferenced map and matched these points to 1950 census tracts. Fourteen of the 21 proposed sites were never built; they fall in 12 census tracts, which I classify as proposed-but-not-built neighborhoods.

I then compare the baseline characteristics of proposed-but-not-built project locations to those of actual public housing locations in Philadelphia. Table 4 compares these two groups and includes all Philadelphia tracts with 1940 data as a citywide benchmark. The initial characteristics of the proposed locations were very different from those of actual public housing locations. Most notably, these proposed-but-not-built project locations had very low initial Black population shares. They were also further from the CBD, less

populous, and had lower unemployment rates. These differences illustrate the dynamics of site selection in one particular city, where whiter neighborhoods often resisted public housing construction. They illustrate why an identification strategy that used proposed-but-not-built project locations as counterfactuals for actually built public housing locations would likely not be credible.

6 The Effect of Public Housing Construction on Neighborhoods

In this section, I analyze the effect of public housing construction on neighborhood demographics and characteristics in neighborhoods that received public housing and in surrounding neighborhoods in subsequent decades. I first describe the research design and then present the empirical estimates.

6.1 Research Design

6.1.1 Recipient Neighborhoods

The ideal experiment to study the effects of public housing construction on neighborhoods would be for public housing authorities to randomly select neighborhoods in each city to receive a public housing project. Then, one could simply compare the recipient and non-recipient neighborhoods over time. However, as shown in previous sections, public housing site selection was far from random. To approximate the ideal experiment, my primary empirical strategy is a series of difference-in-differences designs with matched controls, comparing recipient neighborhoods to neighborhoods that were initially observationally similar on the variables that predicted site selection. This design is similar in spirit to that used by Chetty, Diamond, et al. (2026) to study the effects of HOPE VI demolitions on neighborhoods.

Step 1: Defining the Donor Pool. First, I restrict the potential donor pool to census tracts that never received public housing and are never within one kilometer of a qualifying public housing project, as defined in Section 4. Excluding these surrounding neighborhoods will allow me to separately estimate the geographic spillovers of public housing while reducing concerns that geographic spillovers contaminate the control group.

I then restrict potential donors geographically. I use three main donor-geography rules, which trade off local comparability against observable neighborhood similarity. The same-CBSA specification restricts selected donors to the recipient tract’s metropolitan area. The

same-region specification permits donor tracts throughout the same Census region. Finally, the national specification permits donors from any metropolitan area. I report results using all three donor pools. Same-CBSA controls are more likely to experience similar local economic and policy shocks, while broader donor pools provide more potential controls and generally exhibit greater covariate overlap.

Step 2: Selecting comparable control neighborhoods. I next select neighborhoods that are similar to recipient neighborhoods in the preceding decades using nearest neighbor matching. For each treatment cohort and donor-geography rule, I estimate a logistic regression for treatment status based on pre-existing neighborhood characteristics. I include the main, statistically significant predictors of site selection from Section 5, as well as several other variables that did not show up as significant in the fully saturated site selection model, but are likely related to both public housing placement and subsequent neighborhood change. In particular, I match on log total population, Black population (in inverse hyperbolic sine), Black population share, log median income, log median rent, unemployment rate, labor force participation rate, homeownership rate, and the share of the population under 18, each measured in the two most recent pre-treatment decades. I also match on (time-invariant) redlining status and distance from the central business district. Using the estimated propensity scores from this regression, I select the three census tracts with the closest propensity scores as the controls for each recipient tract, with replacement.¹³ A control may therefore be selected for several different tracts. Matching each cohort separately uses the pre-treatment characteristics relevant to that cohort. For example, for recipient tracts in the 1950 cohort, I estimate propensity scores based on neighborhood characteristics measured in 1940 and 1930.

Step 3: Pooling donors and constructing weights. Within each treatment cohort, I take the union of all distinct controls selected for at least one recipient tract, so that a control tract that was selected multiple times appears only once for that cohort. Recipient tracts are unique to a particular cohort, but control tracts can appear in multiple cohorts.

The matched control neighborhoods resulting from nearest-neighbor matching are more similar to the recipient neighborhoods in terms of observable characteristics, but they are not necessarily perfectly balanced on their pre-treatment characteristics. So, following Chetty, Diamond, et al. (2026), I re-estimate the same logistic regression in the matched sample, again separately for each treatment cohort, to adjust for the remaining differences between the recipient tracts and selected controls. Recipient tracts receive weight one, and control tracts i in treatment cohort g with fitted probability $\hat{p}_{ig}$ receive a

¹³Results are similar when using a different number of controls.

preliminary weight equal to their estimated odds of recipient status:

$$\tilde{w}_{ig} = \frac{\hat{p}_{ig}}{1 - \hat{p}_{ig}}$$

Thus, among the selected controls, greater weight is placed on neighborhoods whose pre-treatment characteristics more closely resemble those of the recipient neighborhoods.

Because my donor selection requires exact matching by donor geography, I rescale the preliminary control weights within each donor-geography cell. Let N_{cg}^R denote the number of recipient tracts in each donor-geography cell c and treatment cohort g , and let $\tilde{W}_{cg}^C$ denote the sum of the preliminary weights assigned to controls in that cell. Each selected control tract then receives the final weight:

$$\frac{N_{cg}^R}{\tilde{W}_{cg}^C} \tilde{w}_{ig}.$$

The normalization factor $N_{cg}^R / \tilde{W}_{cg}^C$ rescales all control weights within a cell so that their total weight equals the number of recipient tracts in that cell. This preserves the distribution of recipient tracts across donor-geography cells, while retaining the relative propensity-score weights of controls within each cell (see e.g., [Iacus, King, and Porro 2012](#)).

Table 5 compares recipient neighborhoods with the full national donor pool, as well as the matched, reweighted controls under each donor-geography rule. Standardized mean differences show the balance between each control group and the recipient tracts. Matching and reweighting substantially reduce differences in neighborhood characteristics, both for variables used in the matching and for those not included. Regional and national controls generally show significantly better balance than the same-CBSA matches. One substantial remaining difference concerns later exposure to urban renewal, which is much more prevalent in recipient tracts compared to matched control areas. This overlap raises the possibility that some estimated neighborhood changes reflect the combined effect of the two programs. I will return to this issue by comparing results for neighborhoods with and without exposure to urban renewal.

Step 4: Stacked Difference-in-Differences Estimation. I then use a weighted stacked difference-in-differences estimator to compare changes in outcomes in the recipient neighborhoods with changes in the reweighted control neighborhoods before and after public housing construction. For each treatment cohort, I construct a panel containing the recipient tracts in that cohort and their selected controls, along with the weights constructed in Step 3, from 1930 to 2010. I assign the controls the treatment year of the corresponding

cohort, and then stack the cohort-specific panels. A control tract may appear in multiple cohort stacks but receive a separate weight and comparison year in each stack.

I then estimate weighted event-study models using the matched samples and weights described above. For most of my outcome variables, I estimate the following linear specifications using weighted least squares:

$$y_{igt} = \alpha_{ig} + \lambda_{gt} + \sum_{\tau \neq -1} \beta_{\tau} (D_{igt}^{\tau} \times R_{ig}) + \varepsilon_{igt}, \quad (2)$$

where y_{igt} is the outcome in tract i assigned to treatment cohort g in decade t , R_{ig} indicates that tract i is a recipient tract in cohort g , and $D_{igt}^{\tau} = \mathbb{1}[t - g = \tau]$ is an indicator for event time, measured in decades relative to the treatment cohort. α_{ig} are tract-by-treatment-cohort fixed effects, which absorb time-invariant differences between recipient and control tracts, while allowing the same control tract to enter more than one treatment cohort stack. λ_{gt} represent treatment cohort-by-decade fixed effects, so that identification comes specifically from comparing differences between recipient and matched control neighborhoods in each decade within each treatment cohort.

I omit $\tau = -1$, the decade immediately preceding public housing construction, so that each β_{τ} measures the differential change in the recipient neighborhoods relative to the reweighted control group since that pre-treatment decade.

The primary outcomes I study are the Black population share, population by race, median income and rent, and homeownership rates. In the main event studies, I estimate the Black population share and homeownership rates in levels, and median income and rent in natural logs, using weighted least squares.

For population by race, some observations are zero, making their logarithms undefined. Log-like transformations such as the inverse hyperbolic sine transformation accommodate zeros, but their estimated treatment effects can depend on the units of measurement and do not generally allow for a percentage interpretation (Chen and Roth 2024; Fitzgerald et al. 2026). I therefore estimate effects on population by race using weighted least squares in levels and Poisson pseudo-maximum likelihood (PPML), both using the matching weights described above. For PPML, I specify the conditional mean as:

$$\mu_{igt} = \exp \left[\alpha_{ig} + \lambda_{gt} + \sum_{\tau \neq -1} \beta_{\tau} (D_{igt}^{\tau} \times R_{ig}) \right] \quad (3)$$

where μ_{igt} denotes conditional mean population. The main population figures report PPML estimates as percentage changes: $100 \times [\exp(\beta_{\tau}) - 1]$.

Identification requires that, in the absence of public housing construction, recipient

neighborhoods and their matched, reweighted controls would have followed parallel trends within each treatment cohort. For the main event studies, I adjust standard errors for spatial correlation following Conley (1999), using a 2-kilometer cutoff.

6.1.2 Nearby Neighborhoods

Public housing construction may also affect neighborhoods that do not themselves contain a project. Indeed, many of the most pressing debates around mid-century public housing concern the extent to which its effects contributed to broader urban change, such as precipitating white flight (Jackson 1985). I define nearby neighborhoods as census tracts that do not contain public housing but lie within one kilometer of a public housing project, as described in Section 4. I estimate the effects on these neighborhoods using two empirical strategies to construct their counterfactuals.

In the first strategy, I consider nearby tracts to be the “treated” tracts and apply the same matching and reweighting procedure used for recipient tracts, described above. Within each treatment cohort and donor-geography specification, I estimate a separate propensity score model for nearby-tract status using each tract’s pre-treatment characteristics. I then select the three nearest controls from the common donor pool, pool the distinct selected controls, re-estimate the propensity score model on the selected sample, and reweight the controls to match the covariate distribution of the nearby tracts. Finally, I estimate the same stacked difference-in-differences specification described above.

In the second strategy, I use the matched control tracts from the recipient-tract analysis in Section 6.1.1 to construct the comparison group for nearby neighborhoods. For each recipient tract, the three matched controls serve as placebo public housing neighborhoods. I identify all tracts within one kilometer of these placebo neighborhoods and exclude any that ever received a qualifying project or were within one kilometer of a project. I then compare the tracts near the actual project with these potential control tracts. I weight the controls using the same propensity-score method described above, giving more weight to controls that more closely resemble the actual nearby tracts before construction. Each actual nearby tract receives a weight of one, and I rescale the control weights so that their total equals the number of actual nearby tracts associated with that recipient tract.

Table 6 and Appendix Table A3 report covariate balance for nearby neighborhoods under the directly matched and placebo-site designs, respectively. In both designs, matched controls are broadly similar to nearby neighborhoods, though balance is generally better in the directly matched design.

6.2 Effects on Recipient Neighborhoods

I now present the estimates of the effect of public housing construction on neighborhood racial and socioeconomic demographics. Overall, I find that public housing construction had dramatic effects on recipient neighborhoods. I first examine the effect of public housing construction on neighborhood racial composition. Throughout this section, the figures report estimates using all three donor pools, while for concreteness, I report magnitudes from the national donor pool specification.

Figure 5 shows the matched event study estimates for the neighborhood Black population share. I find large effects of public housing construction on the subsequent share of Black residents in recipient neighborhoods. At the first census following construction, the estimated increase is 3.1 percentage points relative to the matched control neighborhoods. The effect grows over time, reaching 10.3 percentage points three decades later. The results are similar across donor pools, and the estimated pre-period coefficients are generally flat, although the same-CBSA specification shows a small, statistically significant pre-trend. Given the significantly better pre-treatment balance, I find the same-region and national donor pool results more convincing. To examine the population changes underlying this shift, Figure 6 shows the results on population by race, estimated using both PPML and weighted least squares in levels. These results generally imply that this shift was mainly driven by a relative increase of Black residents. Three decades after construction, the national PPML estimate indicates a statistically significant 42.0% increase in Black population, while the WLS estimates suggest a sizable average increase in the number of Black residents as well. The estimates on white population are negative in the long run, though they are usually statistically insignificant.

I also find that public housing construction produced persistent changes in the economic composition of recipient neighborhoods. Figure 7 shows the effects of public housing on median rent, income, and homeownership rates, showing broad declines in all three relative to their matched controls. I find median income and homeownership rates fall immediately after construction, while median rents do not change initially but decline over the longer run. Three decades after construction, the estimated changes are -14.9% for median income, -14.6% for median rent, and -5.9 percentage points for homeownership.¹⁴

¹⁴The figures display log coefficients for median income and rent. I report their implied percentage changes as $100[\exp(\beta) - 1]$.

6.3 Effects on Nearby Neighborhoods

Figure 8 shows PPML estimates for population by race in nearby neighborhoods. The long-run point estimates indicate declines in total and white population under all three donor pools. The Black-population decline is statistically significant only with same-CBSA controls. The estimated increase in Black population share three decades after construction is 4.7 percentage points with national controls, with similar estimates under the same-CBSA and same-region pools (Figure 9). These changes in racial composition do not require an increase in the number of Black residents.

Figure 10 shows the effects of public housing construction on median rent and income in the nearby neighborhoods. I find declines in median income in nearby neighborhoods, with an estimated immediate change of -2.3% with national donors and similar estimates under the other pools. Three decades later, the income point estimates remain negative under all three donor pools, but none is statistically distinguishable from zero at the 5% level. This may indicate some degree of economic decline or sorting in these nearby neighborhoods. Rent estimates in the directly matched nearby design are not statistically distinguishable from zero.

Directly matched nearby tracts show increases in Black population shares. The placebo-site design produces smaller long-run point estimates that are not statistically distinguishable from zero under any donor pool (Figure A8), so I interpret nearby effects more cautiously than recipient-tract effects. Appendix Figures A7 and A9 report the corresponding population and housing-market estimates.

6.4 Robustness Checks

Figures A10 and A11 report sensitivity to selecting one, two, or three matched controls under each donor-pool rule. Table A4 compares Conley standard errors using 2 km and 5 km distance cutoffs for effects three decades after construction, holding samples, matching weights, and fixed effects constant.

One concern is that public housing construction could affect control neighborhoods through residential relocation, particularly within the same metropolitan area. Figure A2 shows that Black population shares rise after construction in the weighted controls for both recipient and nearby neighborhoods under all three donor pools. The estimated increases therefore reflect faster increases in recipient and nearby neighborhoods, rather than observed declines among their controls. While this pattern does not rule out spillovers, it shows that the results are not driven by falling Black population shares in the control group.

A separate concern is that unobserved shocks occurring between census dates could be correlated with both project site selection and neighborhood outcomes. For example, if a neighborhood experienced an influx of Black residents after the baseline census, this demographic shift could have influenced where public housing was sited. Because the matching is based on baseline census characteristics, it would fail to account for these changes, and the estimates could reflect pre-existing demographic trends rather than the causal effect of public housing. To assess this, I exploit the fact that projects open at different points within the decade. A project that opens shortly after a census is matched on covariates that are nearly contemporaneous, while one that opens just before the next census is matched on covariates that are up to nine years stale. I classify recipient tracts as early openers (5 to 9 years before the next census) or late openers (0 to 4 years before the next census), pooling the 1950, 1960, and 1970 cohorts, and estimate separate event studies for each group. The 1980 cohort is excluded because the project inventory ends in 1973, so every project in that cohort opened 7 to 9 years before the 1980 census and the cohort has no late openers (143 of 1,003 recipient tracts). If changes between census dates confound the estimates, effects may differ for late openers, whose matching characteristics are measured further before construction. Figure A6 compares the estimated effects three decades after construction for early and late openers. With national donors, the effects on the Black population share, median income, and median rent are similar for the two groups, and none of those three contrasts is statistically significant at the 5% level. Total population grows more in the tracts of early openers, and both that contrast and the white-population contrast are significant at that level. This comparison helps assess sensitivity to the timing of project openings, although similar estimates would not by themselves rule out bias.

7 Mechanisms and Heterogeneity

Next, I explore heterogeneity in the effects of public housing construction across several dimensions. In particular, I examine how the effects vary with initial neighborhood conditions, project characteristics, and construction cohort. To estimate heterogeneous effects, I interact treatment with the category of interest. Where the category is also defined for control tracts (e.g., baseline neighborhood characteristics), I classify the matched controls in the same way and include category-by-treatment-cohort-by-decade fixed effects, so that recipient tracts are compared with matched controls in the same category. The controls and their propensity-score weights are the same as those outlined in Section 6.1.1, except that I rescale the control weights so that their total equals the number of recipient tracts in

the same category and treatment cohort. When the category exists only for recipient tracts, such as project characteristics, all categories share the matched controls and weights as in Section 6.1.1. Where the text quotes percentage changes in population counts, they are PPML estimates on the same samples; the figures show weighted least-squares effects in residents.

7.1 Initial Racial Composition

I find little evidence of tipping: white residents left recipient neighborhoods that began with any established Black presence, whether or not the initial share was in the range where tipping models place the threshold, and stayed in nearly all-white ones. Models of tipping predict that a neighborhood's initial racial composition determines whether a demographic shock sets off a broader racial transition ([Schelling 1971](#); [Card, Mas, and J. Rothstein 2008](#)), and historical accounts describe public housing construction coinciding with the rapid turnover of white or integrated neighborhoods into largely Black ones ([R. Rothstein 2017](#); [Jackson 1985](#)). To test for tipping, I divide recipient neighborhoods into three groups by their Black population share in the decade before construction: below 1% ($N = 299$), 1–12% ($N = 210$), and at least 12% ($N = 494$). The 12% cutoff is close to the mean tipping point of 11.9% that Card, Mas, and J. Rothstein (2008) estimate for 1970–80. The same cutoffs classify nearby neighborhoods and donors, so each group is compared with controls of similar initial composition. Figure 11 reports population by race in levels, because the lowest group starts with almost no Black residents; for white population I also quote proportional changes, because the groups begin with very different numbers of white residents.

Using national donors, recipient tracts in the 1–12% group gain 530 Black residents and lose 442 white residents three decades after construction, and tracts with initial Black shares of at least 12% gain 661 and lose 374. The two groups are not distinguishable. Initially Black neighborhoods had fewer white residents to lose, and in proportional terms their white population falls by 23.9%, against 16.1% in the 1–12% group; median income falls by 19.5% and 18.3%. Nearly all-white recipient tracts look different. They gain 500 Black residents, their white population does not fall (it rises by 650 residents in the first decade), their total population grows by 679 residents, and their median income falls by only 4.4%. The white-population and income differences between the lowest group and each of the other two are statistically significant with national and same-CBSA donors; the white losses of the other two groups are themselves significant only with same-CBSA donors.

Nearby tracts lose white residents in all three groups, though only the 1–12% estimate is precise. That group loses 437 white residents with national donors and more with same-CBSA and same-region donors, and the estimate is significant under every donor pool. In proportional terms, however, the loss is 12.0% in the 1–12% group and 22.0% in the highest group, and no proportional difference between nearby groups is statistically significant under any pool.

Appendix Figure A3 shows Black-share effects for recipient tracts. Black share rises in every group: with national donors, by 7.7 percentage points in the lowest group, 12.1 in the 1–12% group, and 11.1 in the highest group. Public housing made recipient neighborhoods more Black wherever they started. What differed was whether white residents left, and the dividing line sits at the bottom of the distribution rather than at the tipping point.

Alternative cutoffs give the same picture. With cutoffs at 5% and 15%, nearby tracts in the middle group ($N = 288$) lose 576 white residents, significant under every pool (Appendix Figure A4), and with terciles of recipient baseline Black share, recipient tracts in the nearly all-white bottom tercile are again the ones that do not lose white residents (Appendix Figure A5).

7.2 Initial Income

The effects of public housing construction were concentrated in neighborhoods that were already poor. I divide recipient neighborhoods by whether their pre-construction median income falls in the bottom tercile of the matched panel's income distribution ($N = 425$ recipient tracts) or above it ($N = 578$), classify nearby neighborhoods and donors with the same cutoff, and compare each group with controls in the same group. Figure 12 shows the estimates three decades after construction with national donors. In low-income recipient neighborhoods, the Black population share rises by 14.2 percentage points, the white population falls by 26.8%, and median income and rent fall by 27.7% and 21.4%. Higher-income recipient neighborhoods also become more Black, with a 6.6 percentage-point rise in Black share, but their white population does not fall and their median income declines by only 3.5%. The differences between the two recipient groups in white population and income are statistically significant under every donor pool, as is the nearby white-population difference. Nearby neighborhoods show the same pattern: around low-income recipient tracts, Black share rises by 7.6 percentage points and white population falls by 21.7%, against 2.4 points and 4.5% around higher-income tracts. Initial income and initial racial composition overlap, since poorer neighborhoods were also Blacker, but only partly: 65% of low-income recipient tracts began with Black shares of at least 12%,

and 56% of those initially Black tracts are in the low-income group.

7.3 Project Size

Larger projects produced larger socioeconomic declines but not larger racial change. I divide recipient neighborhoods into terciles of the number of public housing units built in the tract: fewer than 218 units ($N = 332$), 218–581 units ($N = 335$), and more than 581 units ($N = 336$). Nearby neighborhoods are grouped by terciles of the units of the projects near them. Because project size is a characteristic of recipient neighborhoods rather than of donors, the three groups share the matched controls and weights of the main design. Figure 13 shows the estimates three decades after construction with national donors. The Black-share increase is similar across the three groups, at 10.2, 9.3, and 11.3 percentage points, and none of the differences is statistically significant under any donor pool. The economic effects are not similar. Median income falls by 23.7% in neighborhoods with the largest projects, against 8.0% and 12.2% in the other two groups, and the contrast between the largest group and each of the others is significant under every donor pool; rents fall by 19.2% in the largest group. White population does not line up with size in the same way: it falls by 21.1% in the largest group, an estimate that is imprecise with national donors and significant with same-CBSA donors, and by 432 residents in the smallest group, whereas neighborhoods with medium-sized projects gain 590 residents and their white population does not fall. Nearby neighborhoods around the largest projects show a 29.9% decline in Black population that differs significantly from the other two groups, and a smaller Black-share increase (2.0 percentage points, against 6.5 around the smallest projects), so the largest projects appear to have concentrated Black residents inside the recipient tract rather than in the surrounding neighborhoods.

7.4 Additional Heterogeneity Analyses

Finally, I ask whether the effects varied with when projects were built, and whether the main results are driven by particular subsets of projects: those in New York City, those in neighborhoods also targeted by urban renewal, and those later selected for HOPE VI revitalization.

I first test whether the effects varied with when projects were built, estimating a separate effect for each construction cohort: the 1950 cohort (first qualifying project completed 1941–50, $N = 214$), the 1960 cohort (1951–60, $N = 306$), the 1970 cohort (1961–70, $N = 340$), and the 1980 cohort (1971–73, $N = 143$). Each cohort keeps its own matched controls and weights, and the four treatment paths are estimated jointly. Figure 14 shows the estimates.

The long-run effects are larger for the earlier cohorts. With national donors, Black share rises by 16.3 and 11.5 percentage points in the 1950 and 1960 cohorts, against 5.9 and 8.9 in the 1970 and 1980 cohorts, and median income falls by 29.0% and 18.8%, against 4.0% and 7.5%. The income declines of the 1950 and 1960 cohorts differ significantly from that of the 1970 cohort under every donor pool; the 1980 cohort, the smallest, is estimated less precisely. Rent declines are similar across cohorts. Part of this pattern reflects project size. Recipient tracts in the 1950 and 1960 cohorts received 713 and 773 public housing units on average, against 484 and 202 in the 1970 and 1980 cohorts, and Section 7.3 shows that the largest projects produced the largest income declines. Cohorts also differ in their cities, their projects, and the calendar periods over which the effects accumulate, so these comparisons do not identify the effect of building earlier.

Next, I ask whether the results are driven by New York City. The New York City Housing Authority (NYCHA) was, and remains, the largest housing authority in the country, it has often been thought of as better managed and more successful than other American local housing authorities ([Bloom 2008](#); [Hunt 2018](#)), and New York contributes a large share of the recipient tracts in my sample. Figure 15 shows results separately for New York City and for other cities. The effects are not driven by New York City: recipient neighborhoods elsewhere show increases in Black population share and declines in income and rents of the same character, and with national donors the Black-share increases are not distinguishable between the two groups. The long-run rent decline is larger in New York City, with a contrast that is statistically significant at the 5% level, whereas the income contrast is estimated imprecisely. The neighborhood effects of public housing therefore do not neatly correspond to the perceived success of local housing authorities.

I next ask whether the results are driven by urban renewal. Section 5 showed that recipient neighborhoods were more likely to be targeted by urban renewal, so the estimates could conflate the two programs. Figure 16 presents the results, separately estimating Equation 2 for neighborhoods that were urban-renewal tracts and those that were not. The effects are not driven by urban renewal: using national donors, recipient tracts in both groups show statistically significant long-run increases in Black population share and declines in income and rents, and none of the contrasts between the two groups is statistically significant at the 5% level for the displayed outcomes. Because urban-renewal status can coincide with or follow public housing construction, these comparisons do not separately identify the effects of the two programs, but the persistence of the effects outside urban-renewal tracts shows that the main pattern is not confined to areas where the programs overlapped.

Last, I ask whether the results are driven by the projects that later proved most dis-

tressed. I link projects to the HOPE VI award roster described in Section 4. Figure 17 compares recipient neighborhoods whose projects were later selected for HOPE VI ($N = 174$) with those whose projects were not ($N = 824$). The effects are not confined to the projects later selected for HOPE VI: Black population shares rise by the same amount in both groups, and income and rents fall significantly in both. Neighborhoods with later-HOPE VI projects do show larger long-run declines in income and rents, consistent with the distress that led to their selection.

8 Public Housing and Long-Run Economic Opportunity

I next examine the adult outcomes of children who grew up in recipient and nearby neighborhoods. This exercise connects the neighborhood changes documented above to later economic opportunity. In particular, I ask whether children living in recipient and nearby neighborhoods had different levels of upward mobility and incarceration rates than those in their matched controls. To do so, I use data on upward mobility and adult incarceration rates for low-income children born between 1978 and 1983 from Chetty, Friedman, et al. (2018), as described in Section 4. I merge these outcomes into the matched samples from Section 6, retaining the selected controls and weights from each neighborhood design. Appendix Figure A12 maps upward mobility around Robert Taylor Homes in Chicago, illustrating differences in adult income ranks across nearby census tracts.

I then estimate a cross-sectional regression to test whether recipient neighborhoods and nearby neighborhoods had lower levels of upward mobility than their initially similar matched neighborhoods. For recipient neighborhoods, I use their selected, reweighted donors, whereas for nearby neighborhoods, I report results using both the directly matched controls and the neighborhoods surrounding the matched placebo sites, as described in Section 6.1. I estimate the following model, separately for recipient tracts and nearby tracts:

$$Y_i = \beta R_{ic} + \mu_c + \epsilon_{ic} \quad (4)$$

where Y_i denotes the Opportunity Atlas outcome for tract i , and R_{ic} indicates recipient or nearby status. c represents the comparison set. For the recipient and directly matched nearby tracts, c represents public-housing construction cohorts, with selected donors pooled within each cohort. μ_c is then a cohort fixed effect. For the placebo-site design, each comparison set contains neighborhoods surrounding a recipient tract and neighborhoods surrounding its matched placebo tracts, and μ_c is a fixed effect for that comparison group. I estimate Equation 4 using weighted least squares with the same

weights as described in Section 6.1. Standard errors are again adjusted for spatial correlation within a 2-kilometer radius following Conley (1999).

The baseline specification includes no additional control variables. I also estimate a version in which I control for 1980 tract-level racial and economic characteristics: in particular, Black population share, median income, and unemployment rate. These characteristics are affected by public housing itself and are therefore “bad controls,” so the adjusted estimates describe conditional associations. Their attenuation can indicate a relationship with the previously documented demographic and socioeconomic changes, but does not identify a mediated share or isolate an independent effect of public housing.

Table 7 presents the results by donor geography. Panel A reports adult household income rank and Panel B reports incarceration, and each pair of columns reports specifications with and without 1980 neighborhood controls. I find that recipient neighborhoods had notably lower rates of upward mobility along with higher adult incarceration rates. In the primary specification, recipient neighborhoods have lower mean income ranks by 3.1, 2.6, and 1.6 rank points relative to weighted donors from the same CBSA, the same Census region, and the national pool, respectively. Their incarceration rates are also higher by 1.0, 0.9, and 0.7 percentage points. These differences are statistically significant at the 5% level under all three donor pools. However, after controlling for racial and economic composition, these gaps largely disappear in magnitude and statistical significance, as shown in the specification with controls, consistent with these differences being related to the demographic and economic characteristics of recipient neighborhoods.

By contrast, evidence that the nearby neighborhoods had much worse long-run economic opportunity is weaker and sensitive to the comparison group. The direct matched comparisons show lower adult income ranks and higher incarceration rates, though these differences are statistically significant only at the 5% level for the same-CBSA design, which has the worst initial pre-period balance. These differences shrink and become statistically insignificant after adding controls. The placebo-site income comparisons vary in sign across donor pools. After adjustment, the national comparison shows a positive gap of 0.72 income-rank points ($p = .033$); the same-CBSA and same-region gaps are not statistically significant at the 5% level. None of the adjusted nearby incarceration gaps is significant at that level.

Ultimately, these results suggest that recipient neighborhoods were lower-opportunity neighborhoods for children who grew up there, with lower rates of upward mobility and higher rates of incarceration. Still, the fact that these differences shrink substantially after conditioning on neighborhood racial and economic characteristics in 1980 suggests that this disadvantage is related to these broader neighborhood characteristics. Together with

the evidence above that public housing changed neighborhood composition, this pattern is consistent with demographic and economic changes contributing to long-run opportunity differences. These conditional comparisons cannot establish that channel or rule out other effects of public housing.

9 Conclusion

In this paper, I used newly constructed data to study the 20th-century U.S. public housing program and its effects on the long-run trajectory of neighborhoods. I first documented that public housing projects were systematically targeted toward neighborhoods that were initially poorer and disproportionately Black, consistent with the program's slum-clearance goals and the racial politics of site selection. I then showed that public housing construction had sizable long-term effects on neighborhoods. Compared to initially similar neighborhoods, recipient neighborhoods saw substantial long-run increases in Black population shares and decreases in median incomes and rents. Using directly matched controls, I also find increases in Black population shares in nearby neighborhoods, with little evidence of long-run effects on incomes or rents; the placebo-site estimates are less conclusive.

Recipient neighborhoods also have lower upward mobility for low-income children than their weighted controls. These recipient-tract mobility gaps are not statistically distinguishable from zero after conditioning on neighborhood characteristics in 1980. Evidence of worse outcomes in nearby neighborhoods is limited and sensitive to the comparison group.

These results suggest that public housing both reflected and reinforced existing patterns of segregation, adding quantitative evidence to longstanding debates about the role of federal policy in creating residential segregation ([R. Rothstein 2017](#); [Trounstein 2018](#)). They also help explain the rationale behind modern reforms to public housing, which emphasize mixed-income developments meant to fit within the existing neighborhood fabric rather than large, highly segregated projects.

Still, the fact that I find long-term effects on neighborhoods does not necessarily imply that public housing was worse for its residents than their outside options. Public housing was often oversubscribed, suggesting it may have provided higher-quality, more affordable housing than the available private-market alternatives. A full assessment of the program needs to take into account its effects on residents themselves. In ongoing work, I take up that question by linking individuals to the public housing projects they lived in.

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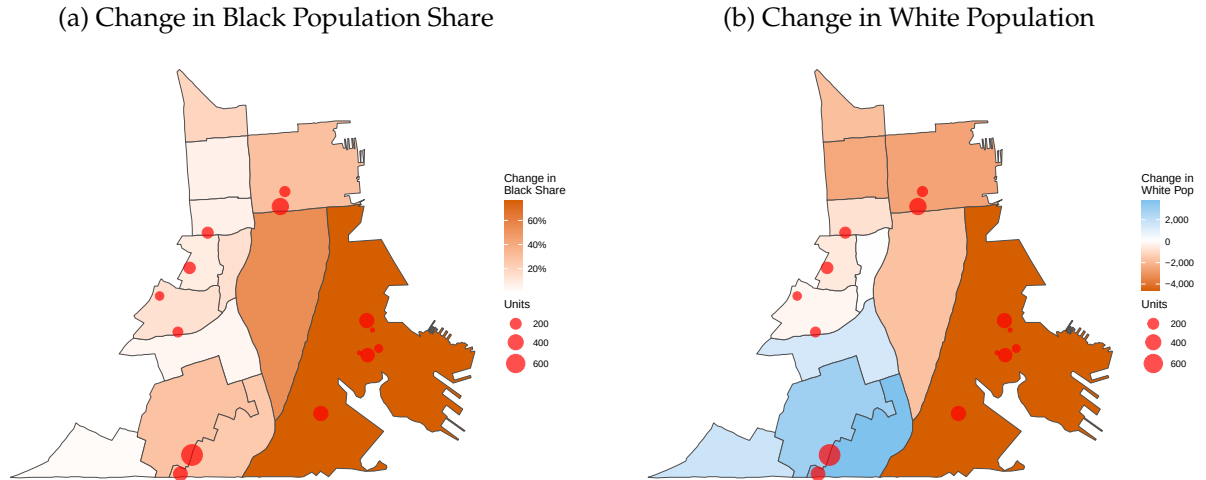
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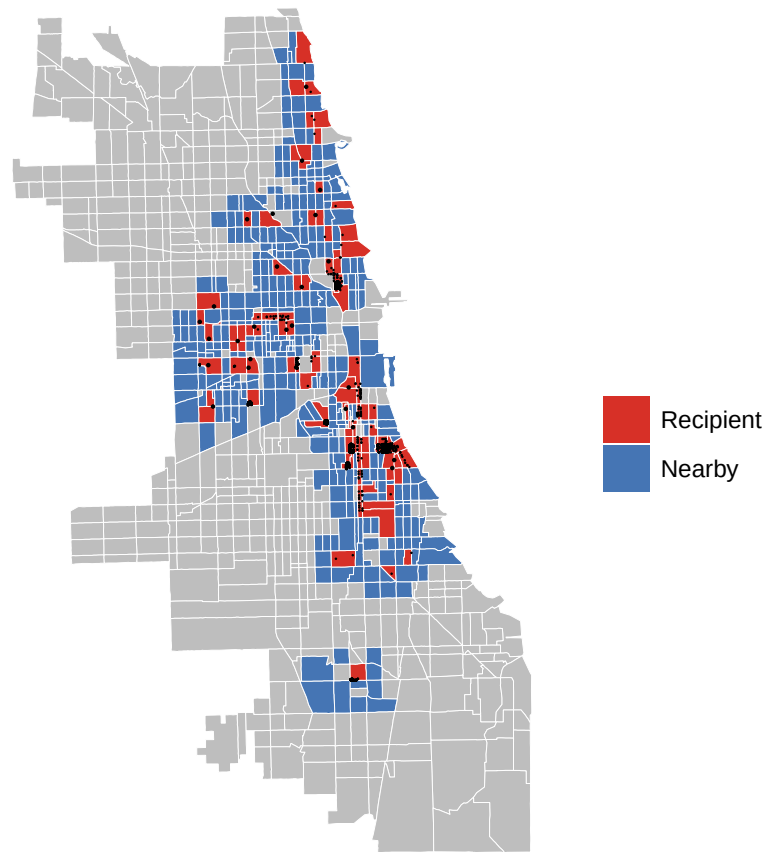
Figures and Tables

Figure 1: Motivating Example: Hunters Point, San Francisco



Note: These maps show demographic changes from 1940 to 1970 in the census tract containing the Hunters Point public housing projects and the surrounding tracts within 2 kilometers (tracts with fewer than 100 residents in 1950 are omitted). Public housing first opened at Hunters Point in 1942–1944; the permanent projects in the inventory, Hunters Point, Hunters View, and Alice Griffith Garden Homes, were completed between 1953 and 1962. Panel (a) shows the change in Black population share; panel (b) shows the change in white population. Red dots mark every public housing project in the inventory located in the mapped tracts, with area proportional to the project’s units. See Section 2 for historical context.

Figure 2: Public Housing Projects and Nearby Areas: Chicago



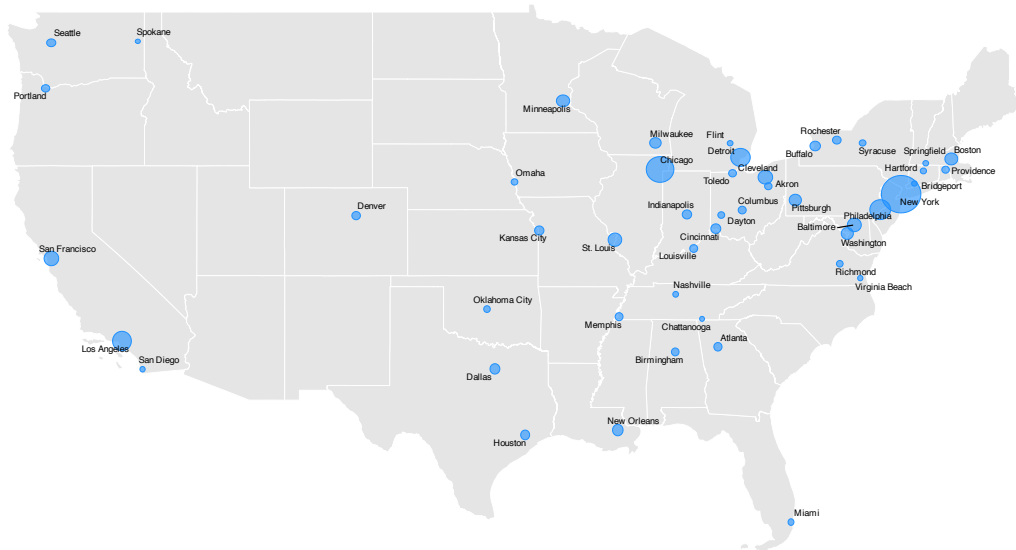
Note: This map shows census tracts in Chicago containing public housing projects built between 1941 and 1973 (red, N=98 tracts) and nearby areas (blue). Spillover areas are defined as census tracts that do not themselves contain a project but lie within 1 kilometer of a qualifying project's buildings. See Section 4 for sample construction details.

Table 1: Sample Attrition

Sample restriction	Tracts	1940 pop (mill.)	CBSAs	Recipient tracts	Projects	Units
Original sample	12,062	38.1	60	1,512	1,506	512,466
Balanced on years (1930-2010)	9,115	34.1	50	1,242	1,199	468,645
Complete variables	8,626	33.2	50	1,218	1,190	464,663
Exclude treated ≤ 1940	8,509	32.6	50	1,101	1,087	414,356
Drop CBSAs <30 tracts	8,459	32.3	48	1,086	1,063	410,323
Drop sparsely populated tracts	7,427	30.8	48	1,003	976	387,905

Notes: Restrictions are applied sequentially. “Balanced on years” requires an observation in every census decade from 1930 to 2010. “Complete variables” additionally requires nonmissing population, Black population share, median family income, median contract rent, labor-force participation, and unemployment in every decade. Tracts first receiving public housing by 1940 are excluded to retain two pre-treatment observations. The CBSA screen requires at least 30 tracts with complete panel data per Core-Based Statistical Area. The final step excludes tracts with fewer than 250 residents in either 1930 or 1940 and geographic outliers exceeding 10 km². Recipient tracts received public housing; projects and units refer to public housing. Population is reported in millions. The final sample contains approximately 23% of the 1940 U.S. population.

Figure 3: Geographic Coverage of CBSAs in Analysis Sample



Note: This map shows the 48 Core-Based Statistical Areas (CBSAs) included in the balanced panel analysis, containing 7,427 census tracts and 1,003 recipient tracts. Each circle is centered on a CBSA, with area proportional to the 1940 population of its sampled tracts. The panel keeps CBSAs with at least 30 tracts with complete panel data; Table 1 lists the full set of sample restrictions.

Table 2: Comparison of Recipient and Non-Recipient Neighborhoods (1940 Baseline)

Variable	Non-recipient	Recipient	Std. Diff.
N (Tracts)	6,424	1,003	
Total Population (asinh)	8.77 (0.72)	9.01 (0.71)	0.336***
Black Share	0.06 (0.17)	0.19 (0.30)	0.533***
Black Population (asinh)	3.23 (2.55)	5.01 (2.98)	0.642***
Median Income (log)	3.01 (0.31)	2.77 (0.32)	0.747***
Median Rent (log)	5.90 (0.40)	5.68 (0.40)	0.550***
Unemployment Rate	0.12 (0.06)	0.15 (0.06)	0.530***
Labor Force Participation Rate	0.56 (0.05)	0.56 (0.05)	0.070**
Distance from CBD (asinh)	7.96 (2.96)	7.10 (3.33)	0.272***
Share Needing Major Repairs	0.07 (0.10)	0.13 (0.14)	0.449***

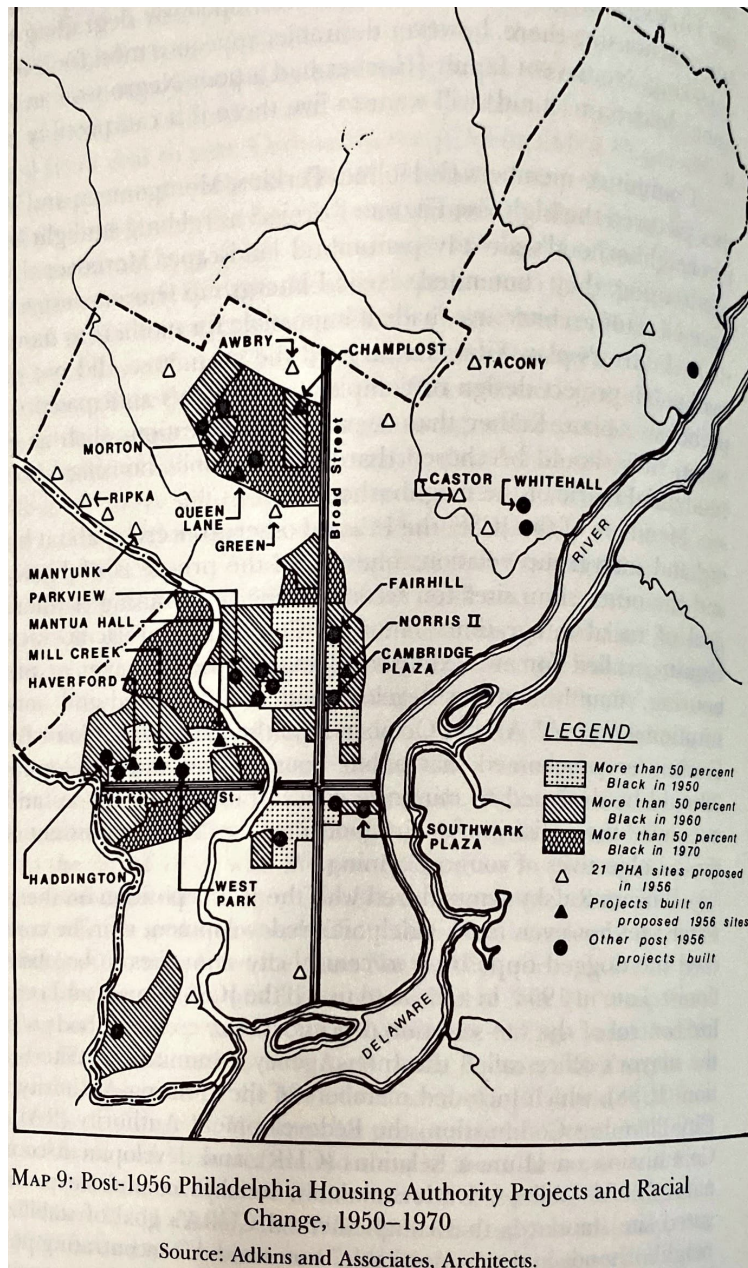
Notes: This table compares 1940 baseline characteristics between census tracts that eventually received public housing projects between 1941 and 1973 and tracts that did not. Sample sizes are reported in the first row. Log denotes the natural logarithm; asinh denotes the inverse hyperbolic sine transformation. Standard deviations are shown in parentheses. The Std. Diff. column reports standardized mean differences (difference in means divided by pooled standard deviation). Statistical significance from two-sample t-tests: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. See Section 5 for analysis of site selection patterns.

Table 3: Public Housing Placement and 1940 Neighborhood Characteristics

	Received public housing, 1941–1973			
	(1)	(2)	(3)	(4)
Black population share	0.257*** (0.038)	0.219*** (0.038)	0.204*** (0.039)	0.209*** (0.038)
Total population (log)	0.036*** (0.010)	0.030*** (0.010)	0.031*** (0.010)	0.029*** (0.010)
Median family income (log)	−0.172*** (0.030)	−0.165*** (0.033)	−0.173*** (0.035)	−0.151*** (0.032)
Median contract rent (log)	−0.030 (0.023)	0.034 (0.033)	0.051 (0.034)	0.037 (0.032)
High-school graduate share		0.123** (0.062)	0.112* (0.064)	0.097 (0.061)
Unemployment rate		0.347** (0.164)	0.362** (0.171)	0.310* (0.160)
Labor-force participation rate		−0.307** (0.137)	−0.289** (0.141)	−0.280** (0.137)
Homeownership rate		−0.201*** (0.039)	−0.185*** (0.040)	−0.189*** (0.040)
Share under age 18		0.688*** (0.151)	0.652*** (0.152)	0.714*** (0.148)
Share age 65 and over		0.037 (0.249)	0.056 (0.256)	0.054 (0.246)
Redlined (HOLC grade D)		0.033* (0.017)	0.034* (0.018)	0.030* (0.018)
Distance from CBD (asinh)		0.004* (0.003)	0.005* (0.003)	0.004* (0.003)
CBD tract		−0.028 (0.027)	−0.043 (0.028)	−0.036 (0.028)
Share needing major repairs			0.089 (0.057)	
Distance to highway (asinh)				0.004 (0.008)
Urban renewal area				0.070*** (0.023)
Observations	7,427	7,427	7,060	7,427
R ²	0.122	0.140	0.138	0.143
Adj. R ²	0.116	0.133	0.131	0.136
CBSA fixed effects	Yes	Yes	Yes	Yes

Notes: This table reports linear probability model estimates of public housing placement on 1940 neighborhood characteristics. The dependent variable equals 1 if a tract received a project between 1941 and 1973 (N=1,003) and 0 otherwise (N=6,424); its sample mean is 13.5%. Income, rent, and population enter in natural logs; distances use the inverse hyperbolic sine transformation. All columns include CBSA fixed effects using the fixed June 2003 delineations described in Section 4.2. Column (2) adds neighborhood controls; Column (3) adds housing disrepair, restricting the sample to tracts with repair data; Column (4) adds urban renewal and highway proximity to Column (2). Standard errors in parentheses allow spatial correlation within 2 kilometers following Conley (1999). Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Figure 4: Historical Map of Proposed and Actual Philadelphia Public Housing Sites, 1956



Note: This map shows the distribution of proposed and actual public housing sites in Philadelphia. Following the 1954 Housing Act, the Philadelphia Housing Authority proposed 21 projects across the city (triangles); 7 were built on the proposed sites (filled triangles) and 14 were canceled after opposition from white communities (open triangles). Circles mark other projects built after 1956. The 14 canceled sites fall in 12 census tracts, the proposed-but-not-built neighborhoods of Table 4. The proposed-but-not-built sites had systematically different baseline characteristics from actual sites (particularly lower Black population shares), illustrating both the political dynamics of site selection and why such sites would not serve as valid counterfactuals in a research design. Source: Bauman (1987), georeferenced by author.

Table 4: 1940 Neighborhood Characteristics: Proposed vs Actual Public Housing Locations in Philadelphia

Characteristic	All tracts	Built sites	Proposed-only sites	Absolute SMD: built vs proposed
Black population share	0.10 (0.19)	0.27 (0.27)	0.01 (0.02)	1.331 ^{***}
Total population (asinh)	8.09 (2.14)	9.21 (1.28)	7.98 (1.78)	0.793 ^{***}
Median family income (log)	2.99 (0.32)	2.85 (0.29)	2.98 (0.27)	0.475
Unemployment rate	0.15 (0.10)	0.20 (0.09)	0.14 (0.05)	0.888 ^{**}
Median contract rent (log)	5.84 (0.43)	5.74 (0.29)	5.81 (0.39)	0.213
Distance from CBD (asinh)	8.75 (2.25)	8.45 (1.88)	9.76 (0.26)	0.970 ^{**}
Redlined (HOLC grade D)	0.21 (0.40)	0.39 (0.49)	0.08 (0.29)	0.751 ^{**}
Urban renewal area	0.17 (0.37)	0.40 (0.49)	0.17 (0.39)	0.532
Tracts	404	57	12	

Notes: This table reports 1940 characteristics of Philadelphia census tracts on 1950 boundaries. The citywide column includes all tracts in the descriptive sample, including built-site and proposed-only tracts; the paper's longitudinal sample restrictions are not imposed. Proposed-only tracts contain proposed-but-not-built sites identified from a 1956 Philadelphia Housing Authority map (Bauman 1987) and no built public housing sites. The first three columns report tract means, with standard deviations in parentheses beneath them; the final row reports tract counts. Statistics use nonmissing observations for each characteristic. Income and rent are in natural logs; population and distance from the central business district (CBD) use the inverse hyperbolic sine (asinh). The final column reports absolute standardized mean differences between built and proposed-only tracts. Stars indicate tests of equality of means for those two groups: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Covariate Balance: Recipient Neighborhoods

Characteristic	Recipient tracts	Eligible controls (SMD)	Matched controls		
			CBSA (SMD)	Region (SMD)	National (SMD)
<i>Matching covariates</i>					
Total population (log)	8.31	8.11 (0.267)	8.24 (0.091)	8.25 (0.082)	8.28 (0.041)
Black population share	30.6	8.9 (0.725)	21.1 (0.284)	26.3 (0.124)	29.0 (0.046)
Black population (asinh)	5.93	3.24 (0.937)	4.98 (0.318)	5.49 (0.147)	5.77 (0.055)
Labor-force participation rate	57.4	56.8 (0.089)	57.7 (0.048)	57.4 (0.007)	57.0 (0.055)
Unemployment rate	11.4	7.0 (0.528)	9.9 (0.156)	11.1 (0.032)	10.5 (0.093)
Median family income (log)	3.09	3.43 (0.806)	3.19 (0.229)	3.13 (0.093)	3.10 (0.027)
Median contract rent (log)	5.67	5.94 (0.786)	5.75 (0.207)	5.68 (0.042)	5.67 (0.003)
Homeownership rate	29.5	48.1 (0.810)	34.6 (0.222)	31.3 (0.085)	32.1 (0.121)
Share under age 18	29.9	26.7 (0.438)	28.6 (0.161)	29.5 (0.054)	29.6 (0.039)
Distance from CBD (asinh)	7.10	8.52 (0.489)	7.32 (0.066)	7.17 (0.021)	6.98 (0.035)
Redlined (HOLC grade D)	32.2	9.2 (0.591)	23.5 (0.196)	27.7 (0.099)	28.7 (0.077)
<i>Additional characteristics</i>					
White population (asinh)	8.17	8.59 (0.331)	8.42 (0.173)	8.30 (0.096)	8.26 (0.067)
Housing needing major repair, 1940	12.9	6.2 (0.582)	11.0 (0.138)	11.3 (0.117)	12.1 (0.055)
High-school graduate share	27.6	39.2 (0.675)	31.9 (0.241)	28.5 (0.053)	28.3 (0.040)
Distance to highway (km)	1.63	2.13 (0.316)	1.70 (0.042)	1.67 (0.027)	1.60 (0.025)
Urban renewal area	25.5	3.0 (0.679)	12.6 (0.333)	10.1 (0.413)	10.9 (0.386)
Tract-cohort observations	1,003	13,766	1,401	1,710	1,785
With 1940 repair data	951	12,958	1,319	1,607	1,682

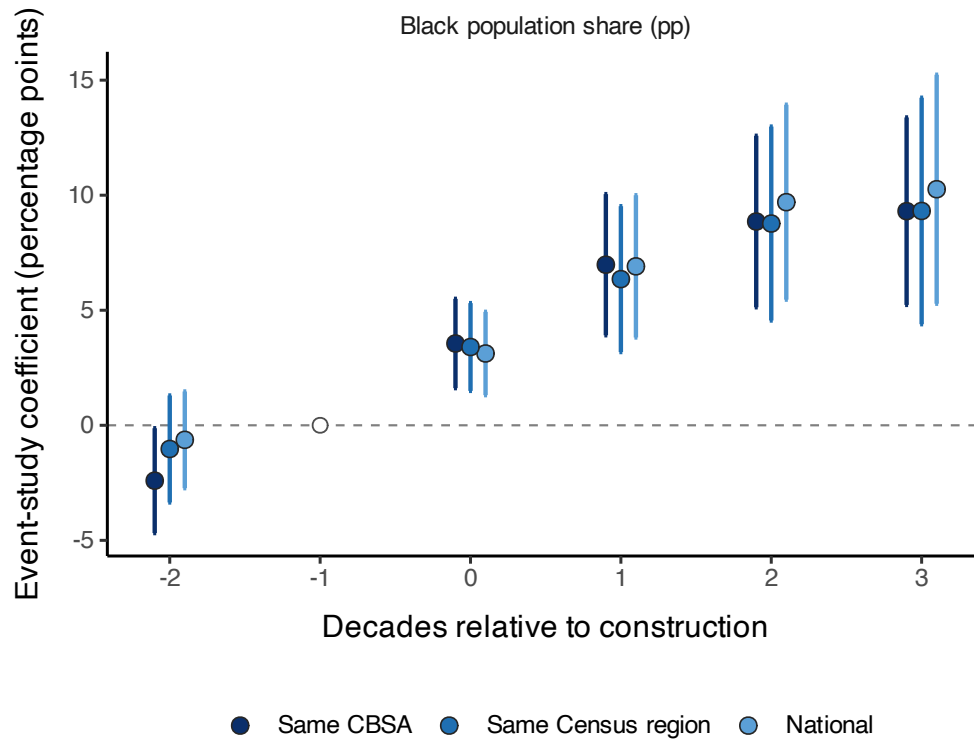
Notes: Means are shown with absolute standardized mean differences (SMDs) from recipients in parentheses. Eligible controls comprise the national donor pool before matching, including selected matches; recipient and nearby tracts are excluded. Matched controls use three nearest neighbors within the same CBSA, Census region, or nationwide, followed by reweighting to improve covariate balance (Section 6.1.1). SMDs use each comparison's weighted pooled standard deviation. Census characteristics use the preceding census, except repair (1940 for every cohort). Shares and rates are percentages; counts are unweighted tract-cohort observations. Income is in thousands of 2000 dollars, rent in 2000 dollars, and distance from the central business district (CBD) in meters before transformation; asinh denotes the inverse hyperbolic sine. Redlined denotes at least 80% of tract area in HOLC grade D. Additional characteristics are excluded from matching. Repair is the share of units reporting condition that needed major repairs; its row excludes missing observations. Highways use the 1996 network; urban renewal denotes at least 5% overlap with historical project areas, which can coincide with or follow treatment.

Table 6: Covariate Balance: Nearby Neighborhoods

Characteristic	Nearby tracts	Eligible controls (SMD)	Matched controls		
			CBSA (SMD)	Region (SMD)	National (SMD)
<i>Matching covariates</i>					
Total population (log)	8.16	8.11 (0.076)	8.14 (0.035)	8.13 (0.048)	8.15 (0.014)
Black population share	16.3	8.9 (0.285)	12.0 (0.160)	14.6 (0.062)	15.6 (0.027)
Black population (asinh)	4.51	3.24 (0.447)	4.03 (0.169)	4.30 (0.073)	4.44 (0.023)
Labor-force participation rate	58.3	56.8 (0.252)	58.4 (0.008)	58.2 (0.013)	58.2 (0.027)
Unemployment rate	11.4	7.0 (0.541)	11.0 (0.047)	11.4 (0.002)	11.2 (0.028)
Median family income (log)	3.17	3.43 (0.635)	3.21 (0.098)	3.18 (0.039)	3.16 (0.007)
Median contract rent (log)	5.79	5.94 (0.445)	5.81 (0.073)	5.79 (0.022)	5.78 (0.028)
Homeownership rate	31.1	48.1 (0.742)	34.2 (0.136)	32.2 (0.049)	32.3 (0.053)
Share under age 18	26.6	26.7 (0.016)	26.2 (0.047)	26.6 (0.005)	26.5 (0.014)
Distance from CBD (asinh)	7.31	8.52 (0.416)	7.39 (0.025)	7.30 (0.003)	7.17 (0.039)
Redlined (HOLC grade D)	21.5	9.2 (0.346)	17.7 (0.097)	19.8 (0.042)	20.4 (0.028)
<i>Additional characteristics</i>					
White population (asinh)	8.44	8.59 (0.142)	8.56 (0.109)	8.49 (0.048)	8.49 (0.048)
Housing needing major repair, 1940	8.9	6.2 (0.273)	8.5 (0.030)	8.9 (0.006)	9.3 (0.033)
High-school graduate share	30.5	39.2 (0.484)	32.6 (0.111)	30.9 (0.018)	31.2 (0.035)
Distance to highway (km)	1.74	2.13 (0.242)	1.82 (0.053)	1.78 (0.028)	1.76 (0.016)
Urban renewal area	16.4	3.0 (0.464)	10.3 (0.181)	8.4 (0.246)	8.6 (0.238)
Tract-cohort observations	2,981	13,766	3,543	4,514	4,707
With 1940 repair data	2,868	12,958	3,415	4,300	4,450

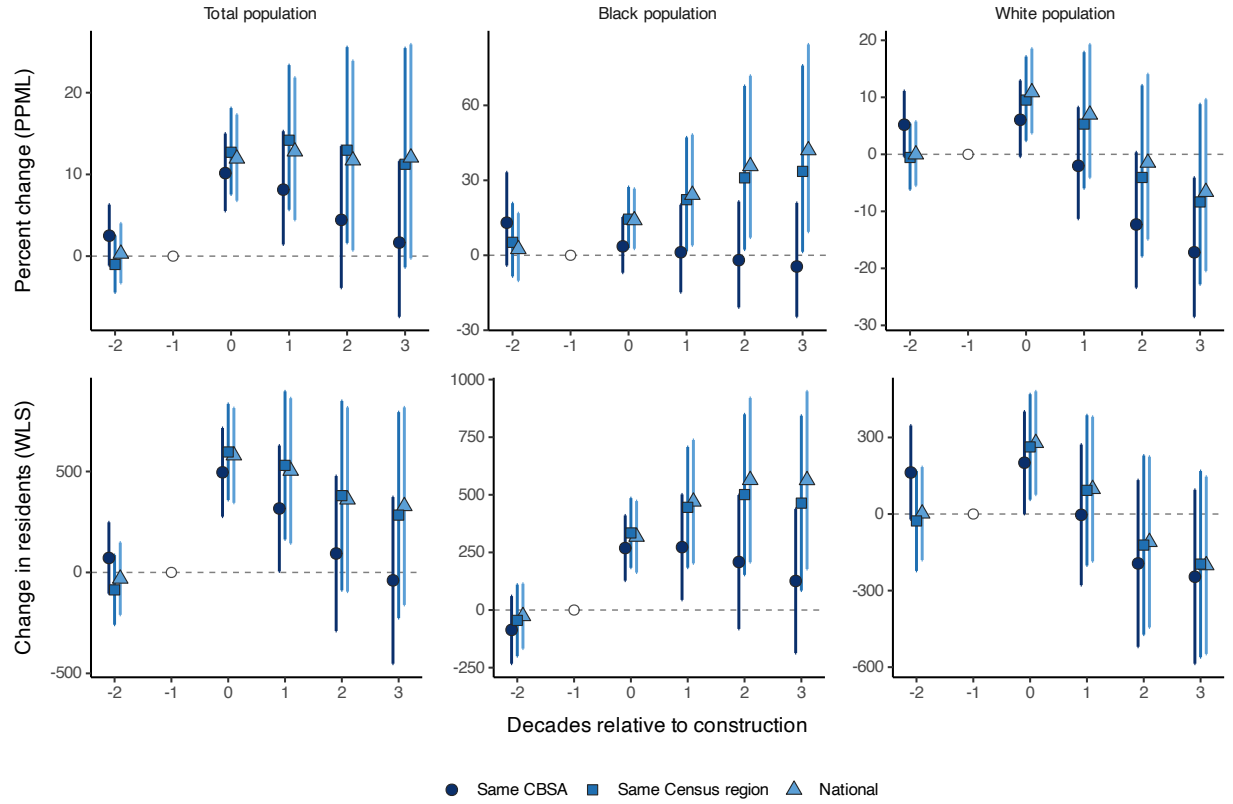
Notes: Means are shown with absolute standardized mean differences (SMDs) from nearby tracts in parentheses. Eligible controls comprise the national donor pool before matching, including selected matches; recipient and nearby tracts are excluded. Matched controls use three nearest neighbors within the same CBSA, Census region, or nationwide, followed by reweighting to improve covariate balance (Section 6.1.1). SMDs use each comparison's weighted pooled standard deviation. Census characteristics use the preceding census, except repair (1940 for every cohort). Shares and rates are percentages; counts are unweighted tract-cohort observations. Income is in thousands of 2000 dollars, rent in 2000 dollars, and distance from the central business district (CBD) in meters before transformation; asinh denotes the inverse hyperbolic sine. Redlined denotes at least 80% of tract area in HOLC grade D. Additional characteristics are excluded from matching. Repair is the share of units reporting condition that needed major repairs; its row excludes missing observations. Highways use the 1996 network; urban renewal denotes at least 5% overlap with historical project areas, which can coincide with or follow treatment.

Figure 5: Effects on Black Population Share in Recipient Neighborhoods



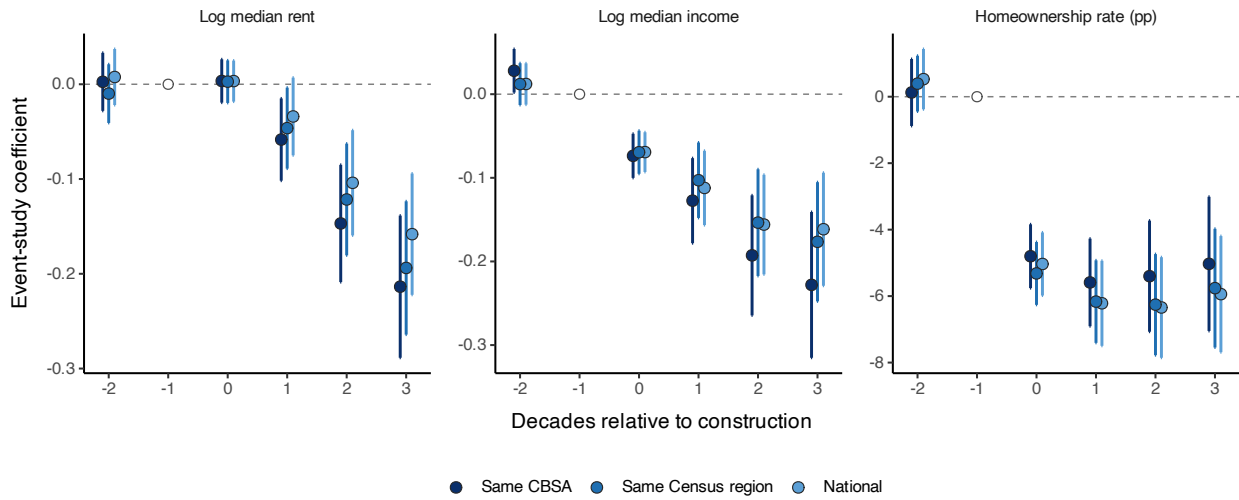
Note: Weighted least-squares event-study estimates of the effect of public housing construction on Black population share in recipient neighborhoods ($N = 1,003$), using Equation 2. Coefficients are percentage point changes. Three controls per recipient tract are matched on pre-construction characteristics, then pooled and reweighted. Same-CBSA controls come from the recipient tract's metropolitan area; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% confidence intervals with Conley standard errors (2 km cutoff). The reference period is $t = -1$, the decade before construction.

Figure 6: Population Effects in Recipient Neighborhoods: Percentages and Levels



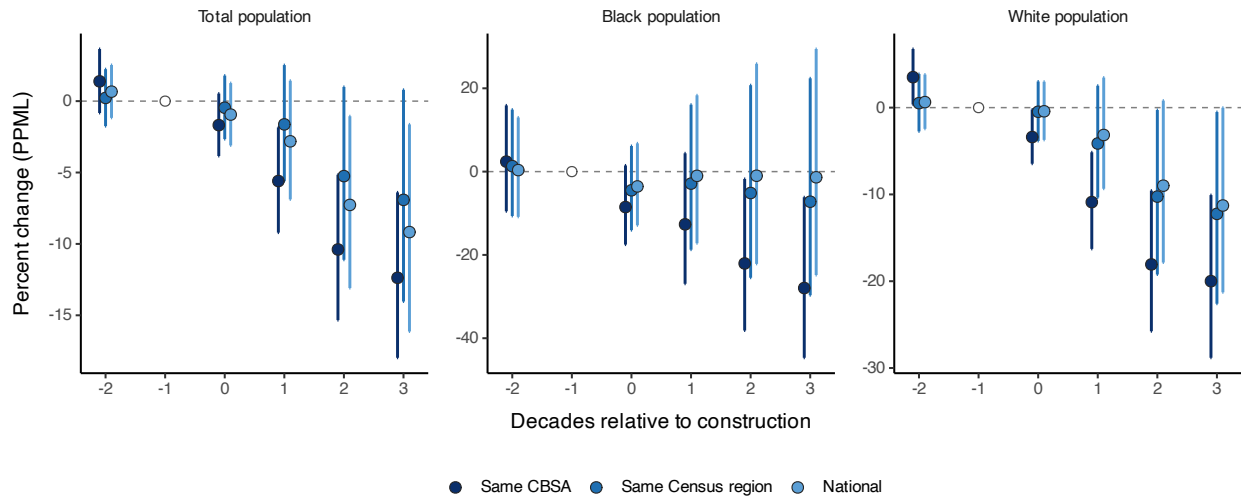
Note: Effects on total, Black, and white population in recipient neighborhoods ($N = 1,003$). The top row reports PPML percentage effects, $100[\exp(\beta) - 1]$, from Equation 3; the bottom row reports weighted least-squares effects in residents from Equation 2. Both start from the same matched samples and matching weights. Three controls per recipient tract are matched on pre-construction characteristics, then pooled and reweighted. Same-CBSA controls come from the recipient tract's metropolitan area; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% Conley confidence intervals (2 km cutoff); PPML interval endpoints are transformed to percentages. Open markers denote the omitted reference decade, $t = -1$.

Figure 7: Economic and Housing Effects in Recipient Neighborhoods



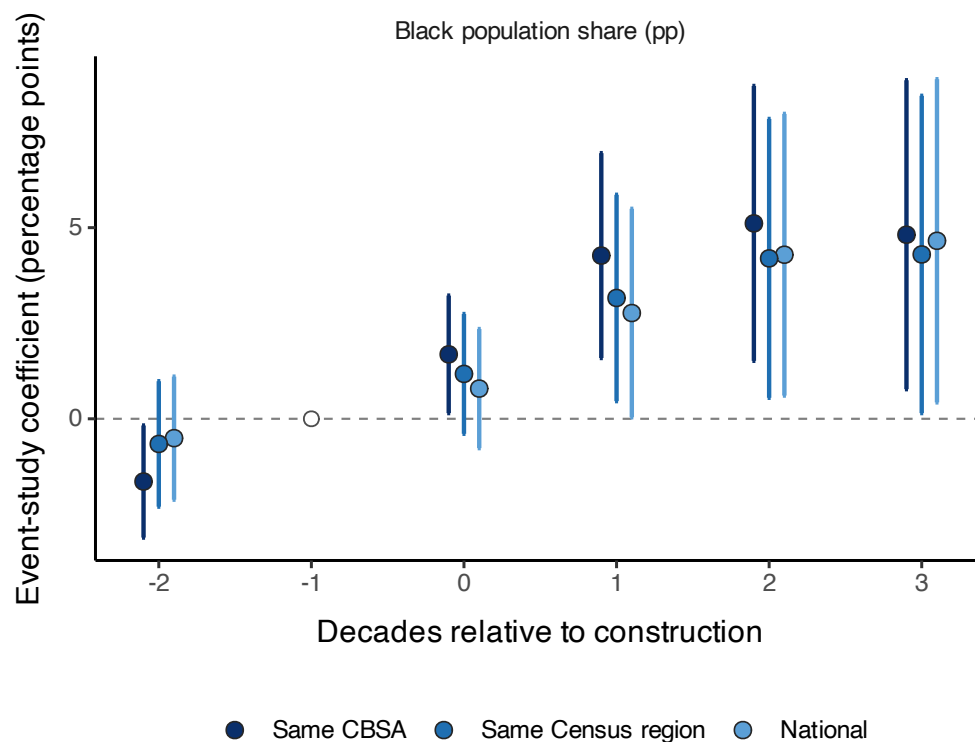
Note: Weighted least-squares event-study estimates for recipient neighborhoods ($N = 1,003$), using Equation 2. Outcomes are log median rent, log median income, and the homeownership rate in percentage points; monetary values are in 2000 dollars before taking logs. Three controls per recipient tract are matched on pre-construction characteristics, then pooled and reweighted. Same-CBSA controls come from the recipient tract's metropolitan area; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% confidence intervals with Conley standard errors (2 km cutoff). The reference period is $t = -1$, the decade before construction.

Figure 8: Spillover Effects: Population and Racial Composition in Nearby Neighborhoods



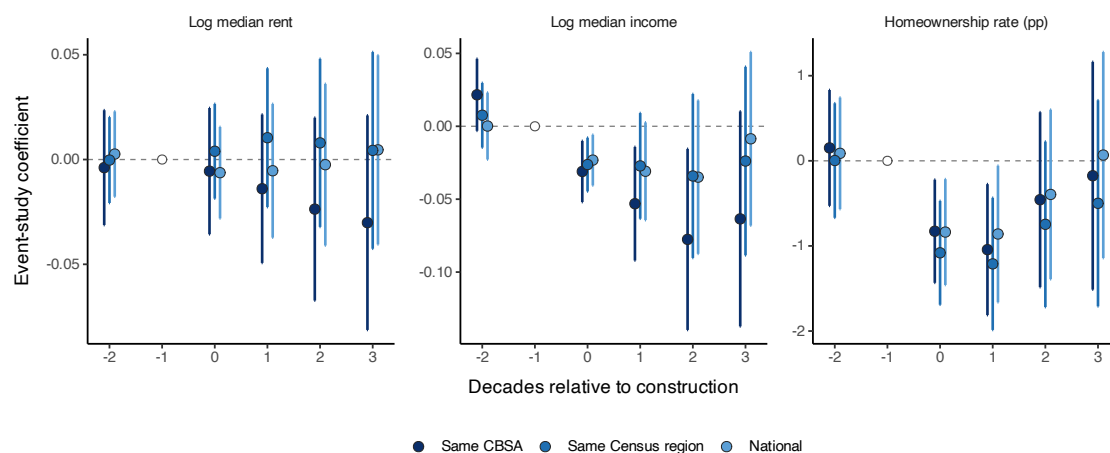
Note: PPML event-study estimates of spillover effects on total, Black, and white population, reported as percentage effects, $100[\exp(\beta) - 1]$, from Equation 3. Nearby neighborhoods contain no qualifying public housing project but lie within 1 km of one ($N = 2,981$). Three controls per nearby tract are matched on pre-construction characteristics, then pooled and reweighted. Same-CBSA controls come from the nearby tract's metropolitan area; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% Conley confidence intervals (2 km cutoff), with endpoints transformed to percentages. The reference period is $t = -1$, the decade before construction.

Figure 9: Spillover Effects: Black Population Share in Nearby Neighborhoods



Note: Weighted least-squares event-study estimates of spillover effects on Black population share, in percentage points, using Equation 2. Nearby neighborhoods contain no qualifying public housing project but lie within 1 km of one ($N = 2,981$). Three controls per nearby tract are matched on pre-construction characteristics, then pooled and reweighted. Same-CBSA controls come from the nearby tract's metropolitan area; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% confidence intervals with Conley standard errors (2 km cutoff). The reference period is $t = -1$, the decade before construction.

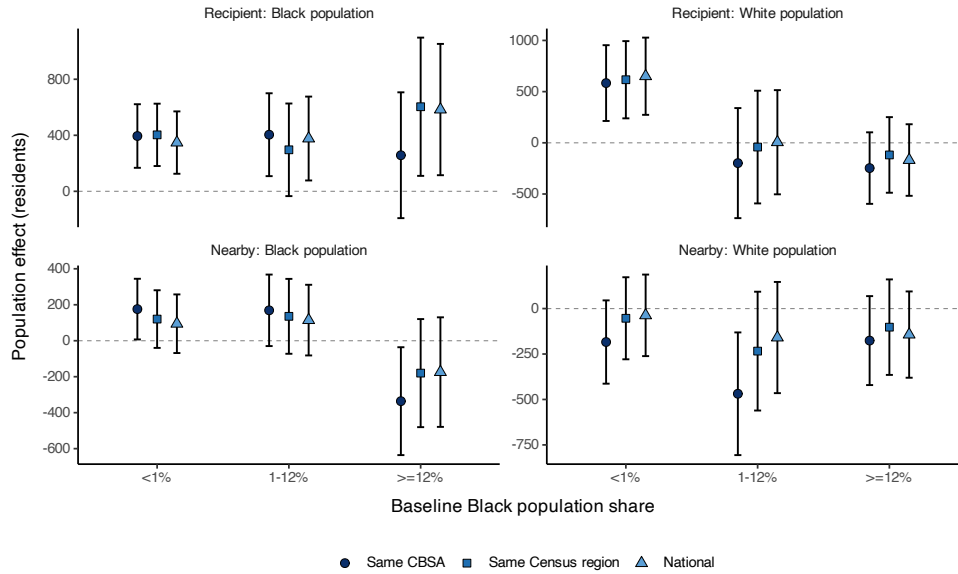
Figure 10: Spillover Effects: Economic and Housing Outcomes in Nearby Neighborhoods



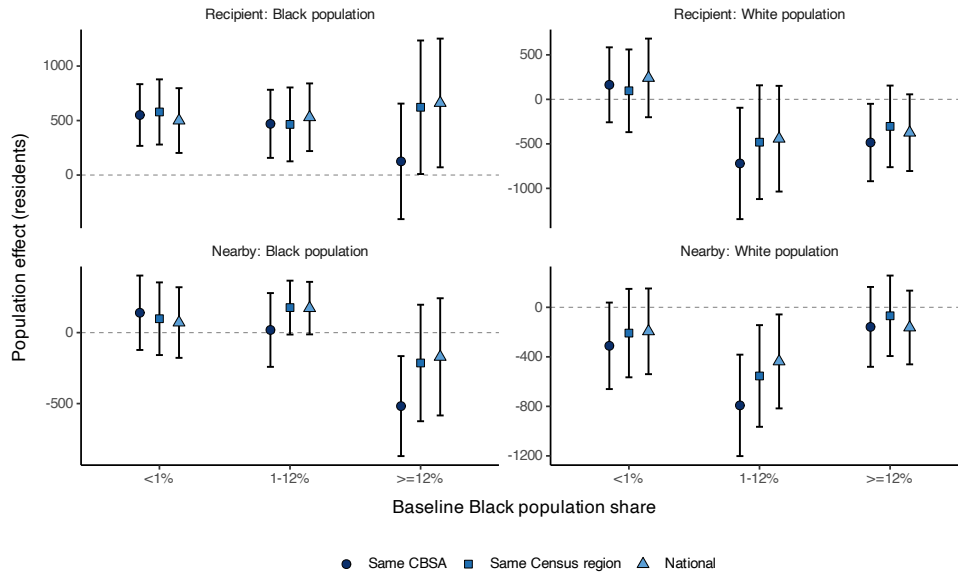
Note: Weighted least-squares event-study estimates of spillover effects, using Equation 2. Outcomes are log median rent, log median income, and the homeownership rate in percentage points; monetary values are in 2000 dollars before taking logs. Nearby neighborhoods contain no qualifying public housing project but lie within 1 km of one ($N = 2,981$). Three controls per nearby tract are matched on pre-construction characteristics, then pooled and reweighted. Same-CBSA controls come from the nearby tract's metropolitan area; same-region controls come from its Census region; national controls may come from any metropolitan area in the sample. Donor pools exclude recipient and nearby tracts. Bars show 95% confidence intervals with Conley standard errors (2 km cutoff). The reference period is $t = -1$, the decade before construction.

Figure 11: Heterogeneity by Baseline Black Population Share

(a) Medium-Run Effects ($t = 1$)

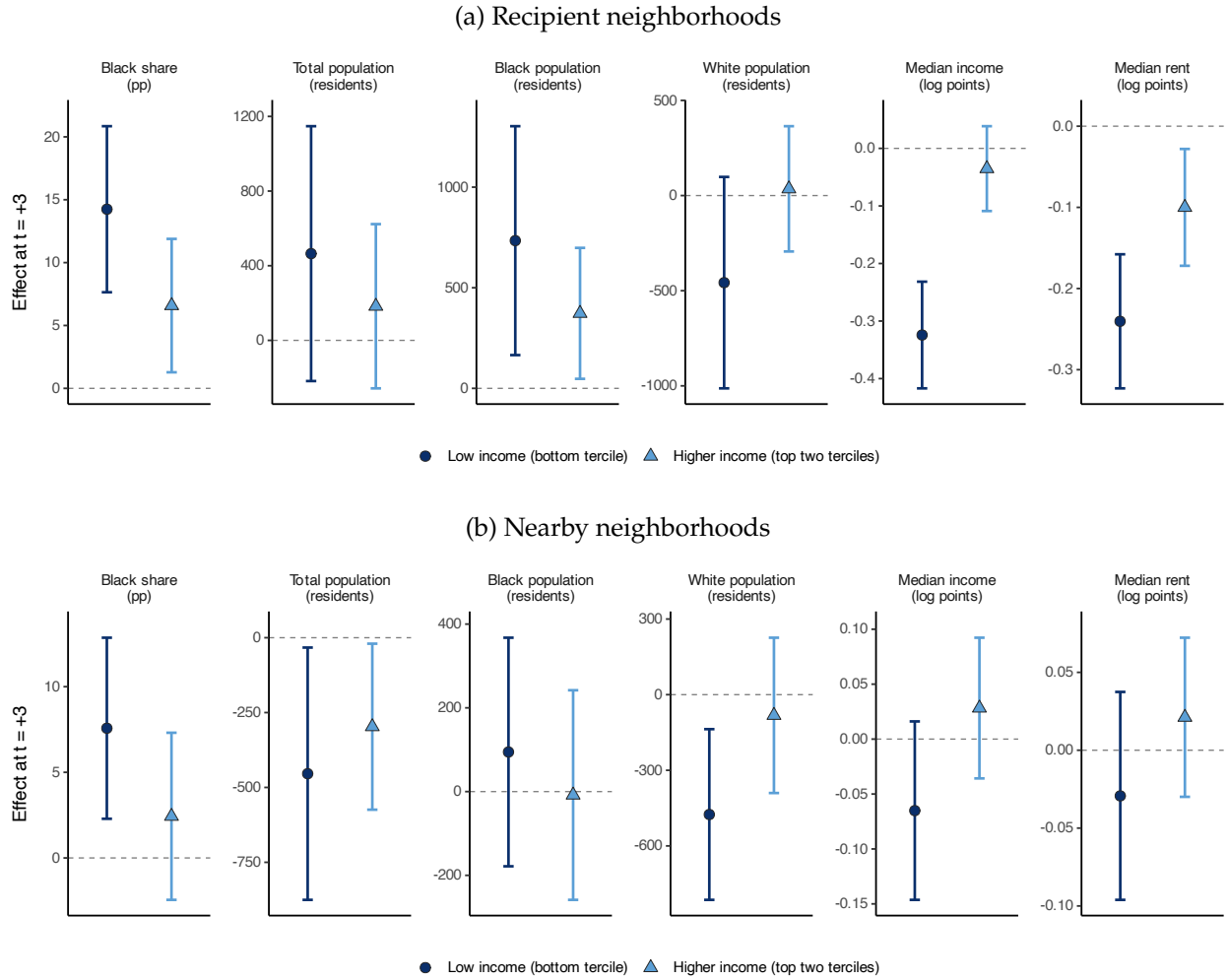


(b) Long-Run Effects ($t = 3$)



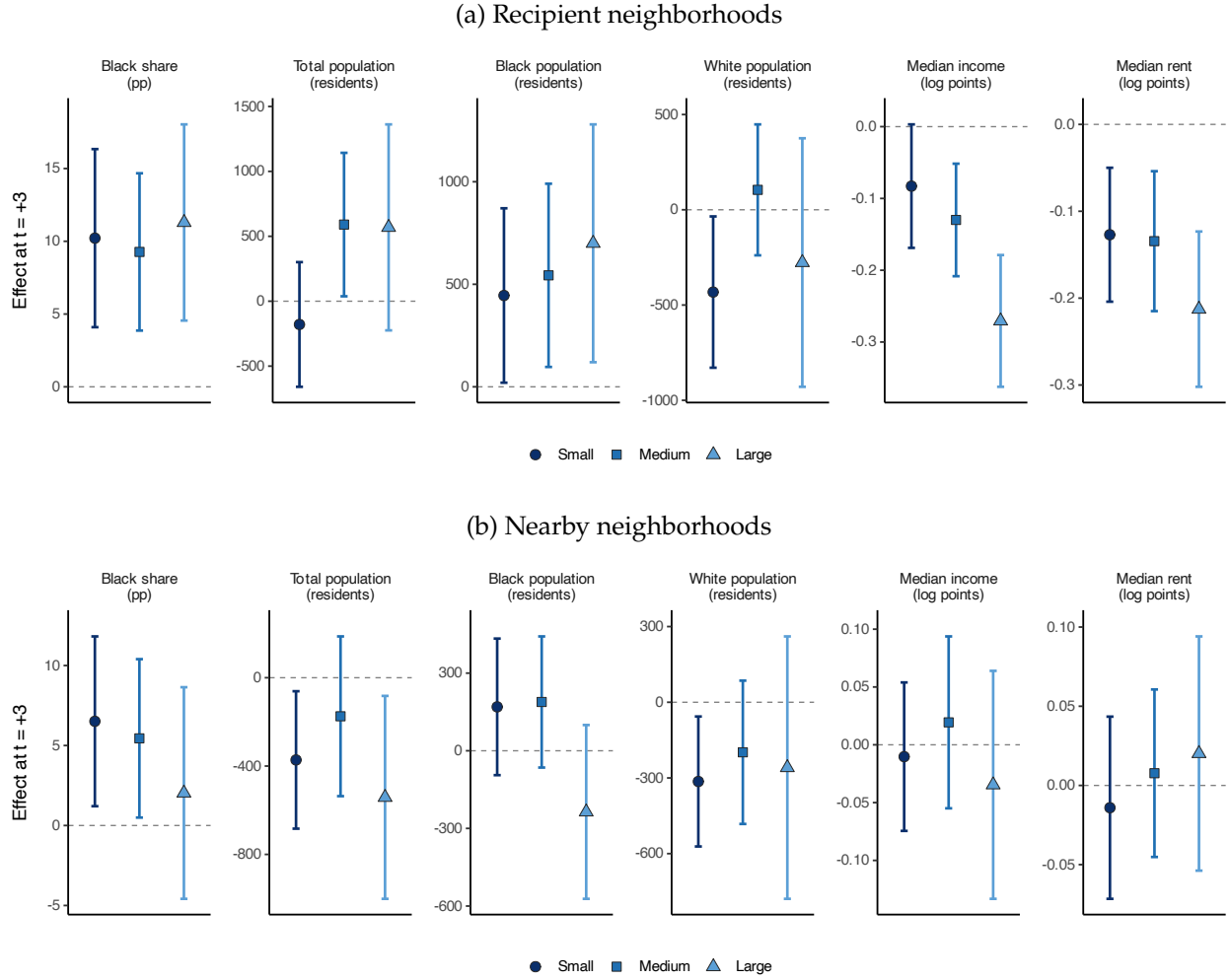
Note: Effects on Black and white population counts, in residents, for recipient and nearby neighborhoods. Baseline Black-share groups are below 1%, 1–12%, and at least 12%; recipient counts are 299, 210, and 494, respectively. Three controls per recipient or nearby tract are initially matched on pre-construction characteristics. Same-CBSA donors are in that tract’s metropolitan area, same-region donors in its Census region, and national donors in any metropolitan area in the sample; recipient and nearby tracts are excluded from donor pools. Comparisons use selected controls in the same baseline-share group and retain the matching weights, rescaled within group and construction cohort. Panels (a) and (b) report weighted least-squares effects one and three decades after construction. Bars show 95% Conley intervals (2 km); the reference decade is $t = -1$.

Figure 12: Heterogeneity by Baseline Income



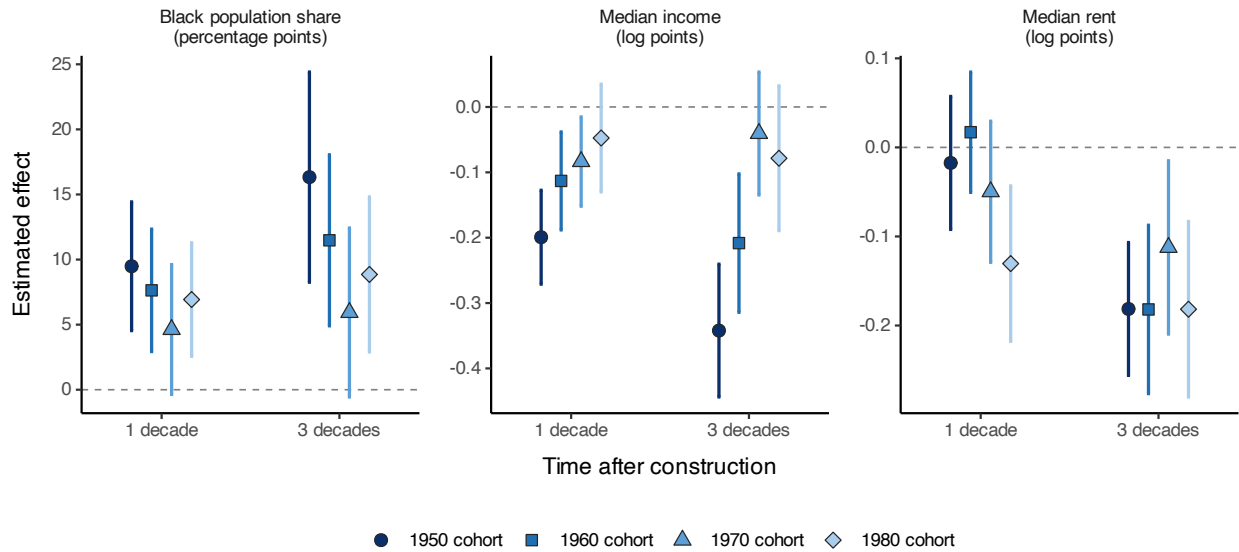
Note: Effects three decades after construction for recipient and nearby neighborhoods by baseline median income. Low-income tracts are in the bottom tercile of pre-construction median income among all tracts in the matched panel (at or below \$21.5 thousand in 2000 dollars with national donors; the cutoff is recomputed within each donor pool); recipient counts are 425 and 578, and nearby counts are 1,194 and 1,787. Three controls per recipient or nearby tract are initially matched on pre-construction characteristics from national donors; recipient and nearby tracts are excluded from donor pools. Comparisons use selected controls in the same income group and retain the matching weights, rescaled within group and construction cohort. Panels report weighted least-squares effects: Black share in percentage points, population counts in residents, and median income and rent in log points. Bars show 95% Conley intervals (2 km); the reference decade is $t = -1$.

Figure 13: Heterogeneity by Project Size



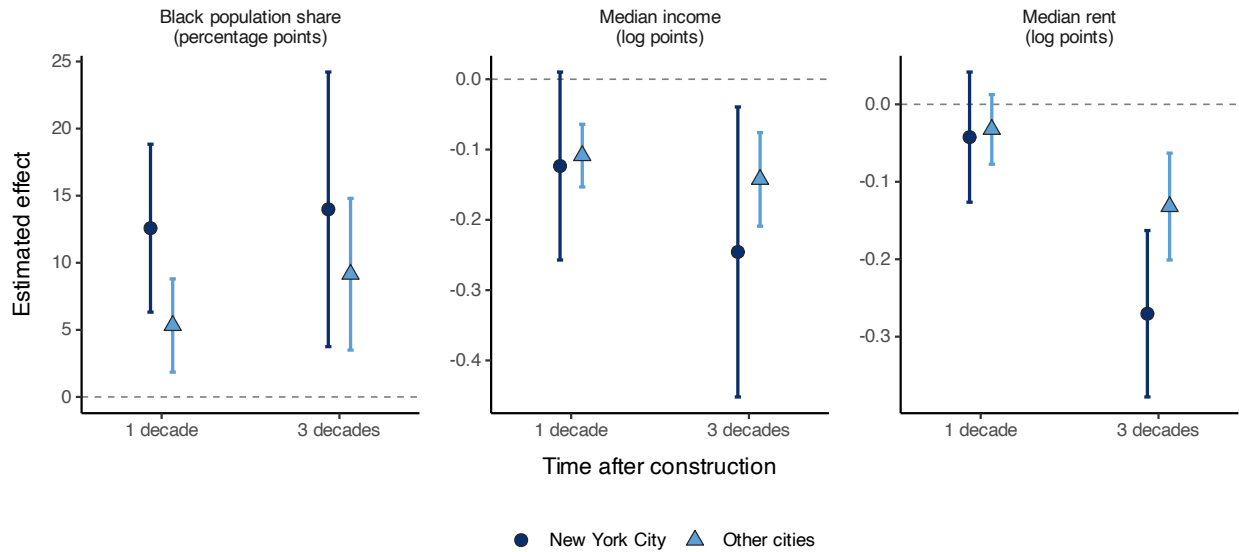
Note: Effects three decades after construction for recipient and nearby neighborhoods by project size. Recipient tracts are grouped into terciles of the total number of public housing units built in the tract (cutoffs at 218 and 581 units; counts 332, 335, and 336), and nearby tracts into terciles of the units of the projects near them (counts 993, 989, and 999). Three controls per recipient or nearby tract are initially matched on pre-construction characteristics from national donors; recipient and nearby tracts are excluded from donor pools. All size groups share the selected controls and matching weights of the main design, rescaled within construction cohort. Panels report weighted least-squares effects: Black share in percentage points, population counts in residents, and median income and rent in log points. Bars show 95% Conley intervals (2 km); the reference decade is $t = -1$.

Figure 14: Heterogeneity by Construction Cohort



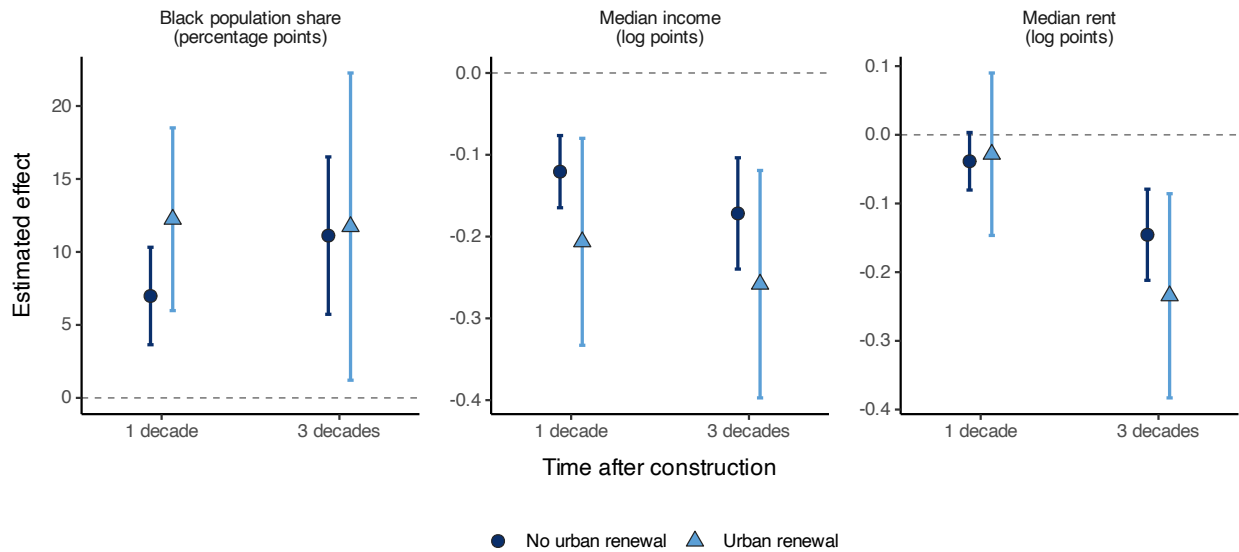
Note: This figure displays separate treatment effects for recipient tracts in each first-treated Census cohort: 1950 (first qualifying project completed 1941 to 1950), 1960 (1951 to 1960), 1970 (1961 to 1970), and 1980 (1971 to 1973). The initial match selects three controls per recipient tract on pre-construction characteristics from eligible tracts across the metropolitan areas in the sample, excluding recipient and nearby tracts. Each cohort retains its own selected controls and matching weights, and the four cohort paths are estimated jointly with tract-by-cohort and cohort-by-year fixed effects. The three panels show Black population share in percentage points and median income and rent in natural logs. Within each panel, points compare effects one and three decades after construction; colors and shapes distinguish cohorts. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). Cohorts differ in places, projects, and calendar periods. See Section 7.4 for interpretation.

Figure 15: New York City vs Other Cities



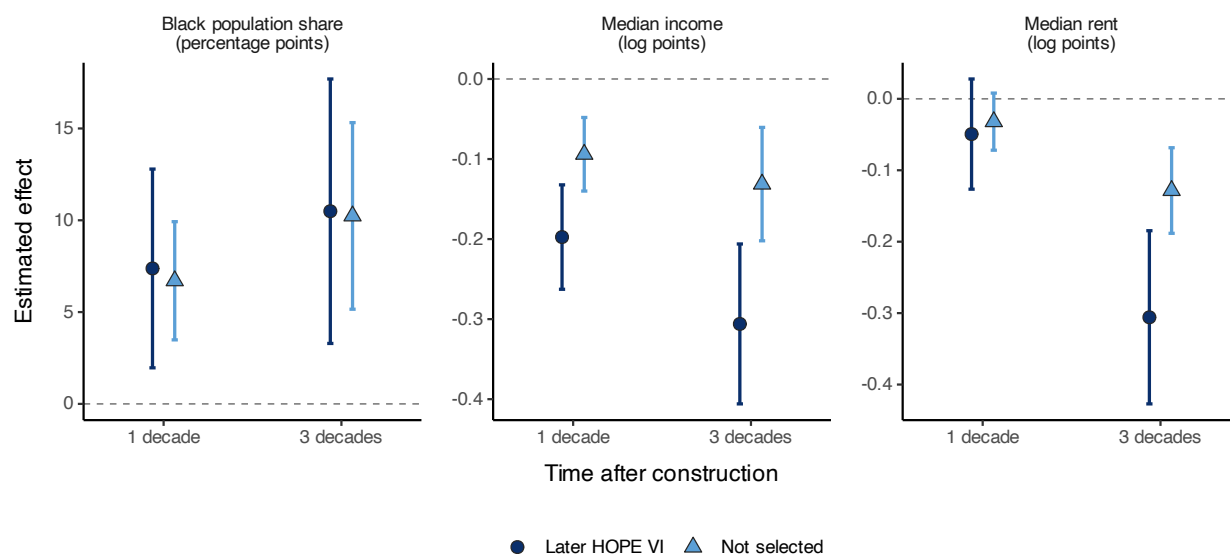
Note: This figure displays heterogeneous treatment effects comparing New York City to other cities in the sample. NYCHA was the largest U.S. housing authority and is often considered better-managed than other PHAs (Bloom 2008). The initial match selects three controls per recipient tract on pre-construction characteristics from eligible tracts across the metropolitan areas in the sample, excluding recipient and nearby tracts. Recipient tracts are compared with selected controls in the same city group (New York City or other cities), retaining matching weights rescaled within group and construction cohort. Estimates follow Equation 2. The three panels show Black population share in percentage points and median income and rent in natural logs. Within each panel, points compare effects one and three decades after construction; colors and shapes distinguish the two subgroups. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.4 for interpretation.

Figure 16: Heterogeneity by Urban Renewal Status



Note: This figure displays heterogeneous treatment effects by urban renewal status, comparing tracts with at least 5% overlap with urban renewal project boundaries to non-urban renewal tracts. This analysis addresses potential confounding between the two programs and tests whether effects differ where both operated. The initial match selects three controls per recipient tract on pre-construction characteristics from eligible tracts across the metropolitan areas in the sample, excluding recipient and nearby tracts. Recipient tracts are compared with selected controls in the same urban-renewal group, retaining matching weights rescaled within group and construction cohort. Estimates follow Equation 2. The three panels show Black population share in percentage points and median income and rent in natural logs. Within each panel, points compare effects one and three decades after construction; colors and shapes distinguish the two subgroups. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.4 for interpretation.

Figure 17: Heterogeneity by Eventual HOPE VI Selection



Note: This figure displays treatment effects by eventual HOPE VI selection, comparing recipient tracts whose projects were later selected for a HOPE VI revitalization grant (174 tracts) with those whose projects were not (824 tracts). Eventual selection is a retrospective marker of project distress, not a treatment, and the comparison is descriptive. The initial match selects three controls per recipient tract on pre-construction characteristics from eligible tracts across the metropolitan areas in the sample, excluding recipient and nearby tracts. Selected controls are pooled and reweighted within construction cohort, and both groups share the same controls. Estimates follow Equation 2. The three panels show Black population share in percentage points and median income and rent in natural logs. Within each panel, points compare effects one and three decades after construction; colors and shapes distinguish the two subgroups. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.4 for interpretation.

Table 7: Long-Run Outcomes in Recipient and Nearby Neighborhoods

	Recipient tracts		Nearby: direct matching		Nearby: placebo-site controls	
	Without	With	Without	With	Without	With
1980 controls						
<i>Panel A: Adult household income rank</i>						
Same CBSA	-3.06*** (0.71)	-0.06 (0.58)	-1.90*** (0.63)	-0.47 (0.47)	-1.09** (0.46)	-0.45* (0.26)
Same Census region	-2.60*** (0.84)	-0.41 (0.53)	-1.35* (0.73)	-0.54 (0.49)	-0.49 (0.51)	-0.32 (0.36)
National	-1.57** (0.74)	0.24 (0.46)	-0.86 (0.67)	-0.26 (0.44)	0.90* (0.51)	0.72** (0.34)
<i>Panel B: Incarceration rate</i>						
Same CBSA	1.03*** (0.26)	0.25 (0.24)	0.50** (0.20)	0.12 (0.17)	0.30** (0.13)	0.15 (0.11)
Same Census region	0.87*** (0.28)	0.29 (0.22)	0.36 (0.22)	0.13 (0.16)	0.05 (0.18)	0.01 (0.14)
National	0.68*** (0.27)	0.22 (0.20)	0.30 (0.21)	0.13 (0.16)	-0.21 (0.18)	-0.15 (0.14)

Notes: Opportunity Atlas outcomes cover children born in 1978–1983, pooling races and genders, with parents at the 25th percentile of national income. Adult household income rank is measured in 2014–2015; incarceration is measured as of April 1, 2010. Income-rank estimates and standard errors are in rank points; incarceration estimates and standard errors are in percentage points. Columns compare recipient tracts with their matched donors; nearby tracts with directly matched donors; and nearby tracts with neighborhoods around matched placebo sites, respectively, using each design’s existing weights. Recipient and directly matched nearby specifications include treatment-cohort fixed effects; placebo-site specifications include recipient comparison-set fixed effects. Standard errors in parentheses allow for spatial correlation following Conley (1999), using a 2-kilometer cutoff and 1950 tract coordinates. 1980 controls are Black population share, median income, and unemployment rate. Adjustment for these post-treatment characteristics is descriptive and does not identify a causal decomposition. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

A Public Housing Data Construction

Section 4 describes the sources I use to construct the public housing dataset used in this paper. This appendix provides additional details on how I construct the historical project inventory, link later building records to projects, screen project locations, and construct project-level population estimates.

A.1 Linking Modern Building Records to Historical Projects

Most historical building records can be attached to projects using the federal project identifiers described in Section 4. Linking the modern building data from the National Geospatial Data Asset to the historical project inventory requires additional checks, because HUD changed its development identifiers with the transition to Asset Management Projects (AMPs) in 2008. Some historical projects received new codes, and some projects could be grouped under a single AMP. A modern development code need not identify the same project as an earlier HUD code.

I link the NGDA data to the historical project inventory in two stages. First, I identify the historical project associated with each modern development. I compare standardized development names with historical project names from the HUD-951 and the 1997 and 2000 Picture of Subsidized Households files, linked to the CDD. I restrict comparisons to the same housing authority, and use locations to assess candidate matches. For developments that remain unresolved, I use a crosswalk that I constructed by matching names and locations between the 2007 and 2008 Picture of Subsidized Households files within each housing authority.

Second, I check which historical project each building belonged to. This step is necessary because identifying the project associated with an AMP does not establish that all of its buildings belonged to that project. I use individual building names and proximity to historical project locations to identify buildings belonging to other projects. When these checks reveal that an AMP contains buildings from multiple historical projects, I reassign buildings with a supported historical identity and leave ambiguous remaining buildings unmatched.

New York City requires a separate linkage. My historical inventory uses the federal, state, and city project identifiers in the 1973 NYCHA *Project Data Book*, while the NYCHA *Address Guide* identifies buildings by development name. I link these records using standardized names and locations, supplemented by reviewed mappings for known aliases and divided developments. I also require that name matches must be geographically con-

sistent with the historical project location. I allow one historical project to link to multiple address clusters when the address guide lists its sections separately.

A.2 Estimating Public Housing Project Populations

This section describes the methodology for estimating the population of each public housing project using the 1977 Picture of Subsidized Households (PSH) dataset. The 1977 PSH dataset reports household-level demographic information for public housing residents, including the number of households in the project, racial composition, household size, income, and age structure. However, it does not provide direct population counts.

To estimate total, white, and Black populations for each project p , I multiply the number of households for group r by the average household size:

$$\text{Population}_{pr} = \text{Households}_{pr} \times \text{Average Household Size}_p.$$

One challenge is that approximately 42% of projects lack household size data. To address this, I impute missing values using two categorical variables that explain substantial variation in household composition: (i) *elderly designation* (all elderly, some elderly, none, or not recorded; projects with a blank designation and those coded not applicable form separate cells) and (ii) *racial composition* (simplified into six categories: all white, all Black, all other race, mixed, no white, and other). For each elderly designation $\times$ racial composition cell, I calculate the mean household size across all projects with non-missing data. Missing household size values are then imputed using the mean of the corresponding cell. This imputation strategy leverages the fact that elderly projects have systematically smaller household sizes, and household size varies by the racial composition of the project.

To validate the imputation strategy, I conducted a hold-out test in which 20% of projects with observed household size were randomly set to missing and then imputed using the cell means calculated from the remaining 80%. The imputation has an R^2 of 0.46 and is essentially unbiased, with a mean error of -0.01 persons.

Table A1: Public Housing Inventory and Analysis Sample

	Assembled inventory	Included	Excluded
<i>A. Sample coverage</i>			
Number of projects	7,649	976	6,673
Share of all inventoried projects (%)	100.0	12.8	87.2
Housing units	974,510	387,905	586,605
Share of all inventoried units (%)	100.0	39.8	60.2
<i>B. Project size</i>			
Mean units per project	127.5	397.4	87.9
Median units per project	60	234	50
Fewer than 50 units (%)	39.0	0.0	44.7
50–199 units (%)	44.5	40.3	45.1
200–499 units (%)	12.2	35.2	8.9
500–999 units (%)	2.9	15.8	1.1
1,000 or more units (%)	1.4	8.7	0.3
<i>C. Completion period</i>			
1935–1940 (%)	2.0	0.0	2.2
1941–1950 (%)	5.8	15.6	4.4
1951–1960 (%)	26.0	28.3	25.6
1961–1970 (%)	46.4	38.1	47.6
1971–1973 (%)	19.9	18.0	20.1
<i>D. Census region</i>			
Northeast (%)	19.9	37.6	17.3
Midwest (%)	20.9	27.9	19.9
South (%)	49.8	21.9	53.9
West (%)	9.4	12.6	8.9

Notes: The assembled inventory combines geolocated projects from the 1973 Consolidated Development Directory with digitized local inventories. It covers completed projects through 1973 and excludes leased housing and Indian housing programs. Included projects are those linked to recipient tracts in the balanced analysis sample, including later construction in already-treated tracts. Excluded projects comprise the remainder of the assembled inventory. Exclusion reflects all project and neighborhood restrictions described in the text. Project-size, completion-period, and regional percentages are shares of projects within each column. Size statistics omit 3 projects with missing unit counts, all excluded from the analysis sample; housing-unit totals sum observed values. A project's full unit count is counted once even if its buildings span several tracts. Regions follow the Census classification, with the District of Columbia in the South.

B Census Tract Data Construction and Harmonization

B.1 Data Sources and Variable Availability

I combine published Census tabulations distributed by NHGIS ([Manson et al. 2022](#)) with IPUMS full-count Census microdata for 1930 and 1940 ([Ruggles et al. 2022](#)). The published tabulations provide the primary inputs for 1950–2000. For 2010, I use decennial Census counts for population, age, and housing tenure, and 2006–2010 American Community Survey (ACS) estimates for income, rent, home value, employment, and education. The ACS measures therefore describe a five-year period, rather than a single census date.

Published tract coverage is more limited in 1930 and 1940. I aggregate full-count microdata to enumeration districts. The Urban Transition Historical GIS Project ([J. R. Logan et al. 2024](#)) provides boundary shapefiles for these districts in 1930 and 1940. I construct the crosswalk to 1950 NHGIS tracts using the share of each district’s area within each tract (Section B.2). These data serve two purposes. First, they add tract-years absent from the published population tabulations. Second, within tract-years already covered by the published data, they fill missing income, rent, home value, labor-force participation, unemployment, age-share, and tenure measures. For these variables, the choice is made separately for each tract-year: I retain the published-data value when it is available and otherwise use the full-count measure. Full-count data also supply the early income measures that are absent from the published tract tables.

Table A2 distinguishes the underlying sources from the format of the inputs. Population counts and the counts underlying shares are additive across areas. Income, rent, and home value are represented by grouped frequency distributions; I allocate the counts in each bin to 1950 boundaries before calculating the median, as described in Section B.2.1.

Income definitions change across these sources. In 1930, the Census did not collect income, so I construct a family-level occupational-income proxy by summing, across family members, the machine-learning-adjusted occupational income scores of Saavedra and Twinam ([2020](#)). In 1940, I sum family members’ reported wage and salary income; this measure excludes the amount of nonwage income. The 1950 published distribution covers families and unrelated individuals and excludes income not reported. From 1960 onward, I use family-income distributions. Thus, the early income measures differ both in what they capture and in the units over which the median is calculated.

Age shares distinguish residents younger than 18, ages 18–64, and ages 65 and older. Full-count records and published tables for 1970–2010 identify the under-18 population directly. Published tables for 1940–1960 combine ages 15–19, so I approximate the under-

Table A2: Sources and Formats of Census Tract Characteristics

Characteristic	Source				Input format
	1930	1940	1950–2000	2010	
Population by race	Both	Both	Census	Census	Counts
Median income	Microdata	Microdata	Census	ACS	Grouped counts
Median rent	Both	Both	Census	ACS	Grouped counts
Median home value	Both	Both	Census	ACS	Grouped counts
Age structure	Microdata	Both	Census	Census	Counts
Homeownership	Microdata	Both	Census	Census	Counts
Labor-market rates	Microdata	Both	Census	ACS	Counts
Educational attainment	—	Both	Census	ACS	Counts

Notes: Census denotes published decennial Census tabulations; Microdata denotes IPUMS full-count records aggregated to tracts. Both indicates that published tabulations and full-count records contribute to the early-year data, with the source-combination rule described in the text. ACS denotes 2006–2010 five-year estimates. Labor-market rates are labor-force participation and unemployment; educational attainment refers to completed schooling, which is unavailable in 1930.

18 population by adding three-fifths of that category to the population younger than 15, assuming equal population at each age within the category.

B.2 Tract Boundary Harmonization

I allocate each source tract or enumeration district to 1950 tracts in proportion to its area of overlap. Before calculating areas, I combine all polygon records belonging to the same source unit. For enumeration districts, the source unit is identified by state, city, census year, and enumeration-district code. This ensures that a district represented by several polygons contributes its count only once. I measure both source areas and overlaps in a common equal-area projection. Figure A1 illustrates the resulting weights for two Chicago examples.

Let A_i denote the full area of source unit i , before restricting it to the 1950 tract map, and let a_{ij} denote its overlap with 1950 tract j . The allocation weight is

$$w_{ij} = \frac{a_{ij}}{A_i}.$$

If the 1950 tract map covers only part of a source unit, its weights sum to less than one. I allocate only the share of its count corresponding to the covered area; the share outside the map remains unallocated. For example, a source tract with 40% of its area inside the 1950 map contributes 40% of its count. Source units with no overlap contribute nothing.

For count variables such as population, housing units, and employment, I sum contri-

butions from the source units overlapping each 1950 tract:

$$X_j = \sum_{i \in I_j} w_{ij} \cdot X_i$$

where X_i is the count in source unit i and I_j is the set of source units overlapping 1950 tract j . For rates and shares, I allocate the numerator and denominator counts separately and calculate their ratio after aggregation. I apply the same weights to the counts in each income, rent, or home-value category before calculating medians, as described below. This method assumes that the counted population or housing units are uniformly distributed within each source unit, which may introduce measurement error in areas where they are spatially concentrated. For instance, if a source tract contains a spatially concentrated Black population but overlaps evenly with multiple 1950 tracts, the method would allocate residents uniformly across those tracts, potentially attenuating spatial estimates of segregation.

B.2.1 Calculating Medians from Categorical Distributions

I construct medians after harmonizing the underlying frequency distributions to 1950 tract boundaries (Section B.2). For each income, rent, or home-value bin, I allocate source-area counts using the geographic crosswalk weights and sum the allocated counts within each destination tract. I then calculate the median from this tract-level distribution, rather than averaging the medians of the source areas.

Let n_i be the count in bin i , with bounds $[L_i, U_i)$, and let $N = \sum_i n_i$. The median bin m is the first bin for which $\sum_{i \leq m} n_i \geq N/2$. I linearly interpolate within that bin:

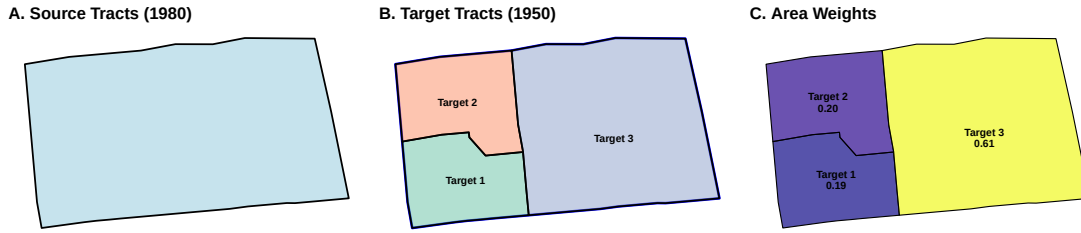
$$\widehat{\text{Median}} = L_m + \frac{N/2 - \sum_{i < m} n_i}{n_m} (U_m - L_m).$$

This calculation assumes a uniform distribution within the median bin. If the median lies in an open-ended top category, I use that category's lower bound. Categories without a monetary value, such as income not reported or no cash rent, are excluded from the calculation and its denominator. For income, the relevant units follow the definitions above; for rent and home value, they are the housing units represented in the respective distribution.

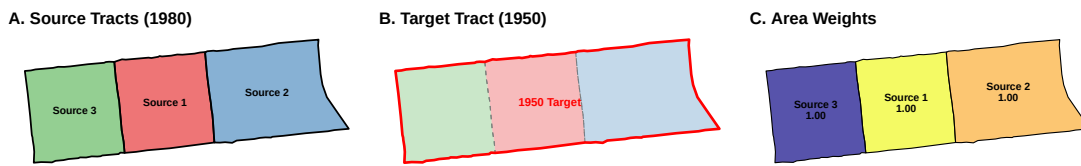
After harmonization and median construction, I convert monetary values to 2000 dollars using the Consumer Price Index from Officer and Williamson (2025). For the 1930 occupational-income proxy, which is expressed in 1950 dollars, I apply the 1950-to-2000 deflator.

Figure A1: Census Tract Harmonization Examples

(a) Split Pattern: One 1980 tract → Multiple 1950 tracts



(b) Merge Pattern: Multiple 1980 tracts → One 1950 tract



Note: These figures demonstrate the area-reweighting methodology for harmonizing 1980 census tracts to 1950 boundaries in a particular example in Chicago. Panel (a) shows a split pattern where one 1980 source tract is divided across multiple 1950 target tracts, with weights representing the proportion of area allocated to each target. Panel (b) shows a merge pattern where multiple 1980 source tracts contribute to a single 1950 target tract.

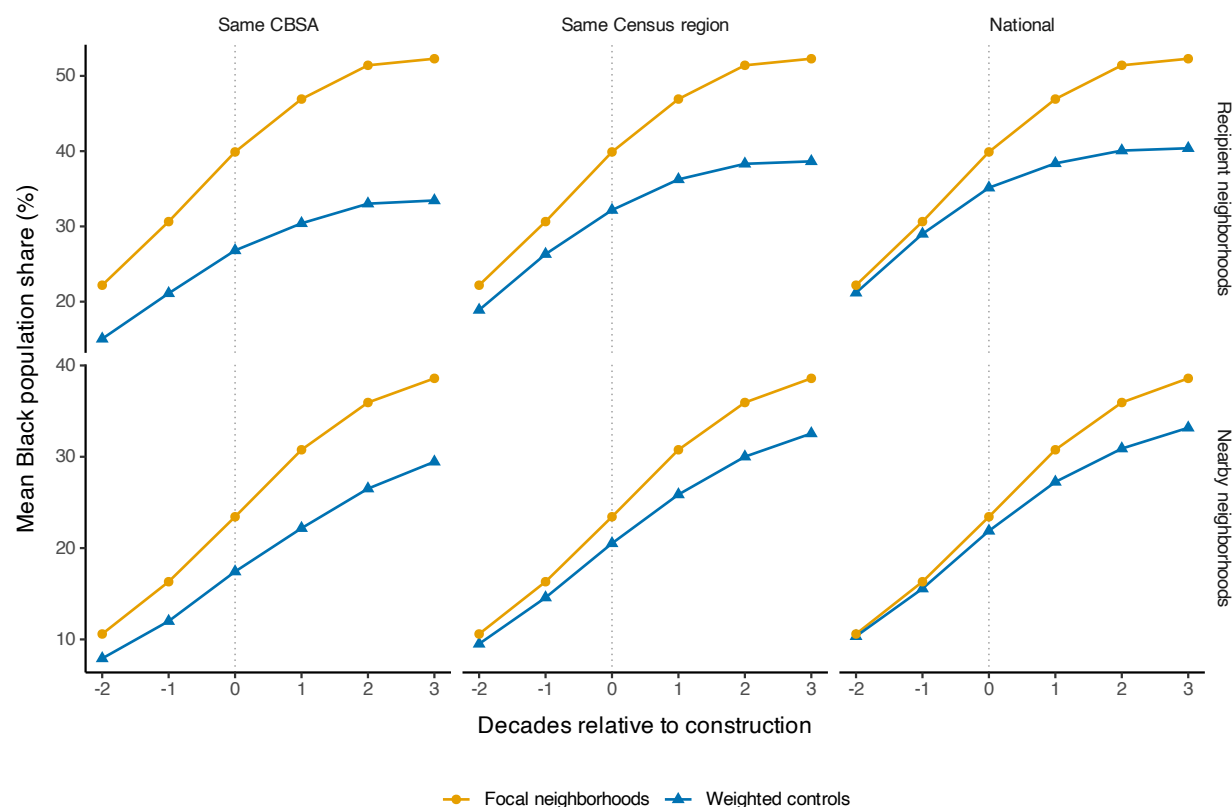
B.3 Opportunity Atlas Outcomes

I harmonize Opportunity Atlas outcomes from 2010 to 1950 tract boundaries using the same source-area overlap shares. Within each destination tract, I average the source-tract outcomes with weights proportional to the overlap share multiplied by the Atlas count of children with parental income below the median. The outcomes are evaluated at parental income percentile 25; this evaluation point differs from the below-median population used for weighting. I retain the supplied outcome estimates, including negative fitted incarceration rates. These crosswalk weights construct the tract outcomes; the neighborhood comparisons then apply the matched-sample weights described in Section 8.

C Supplementary Results and Robustness Checks

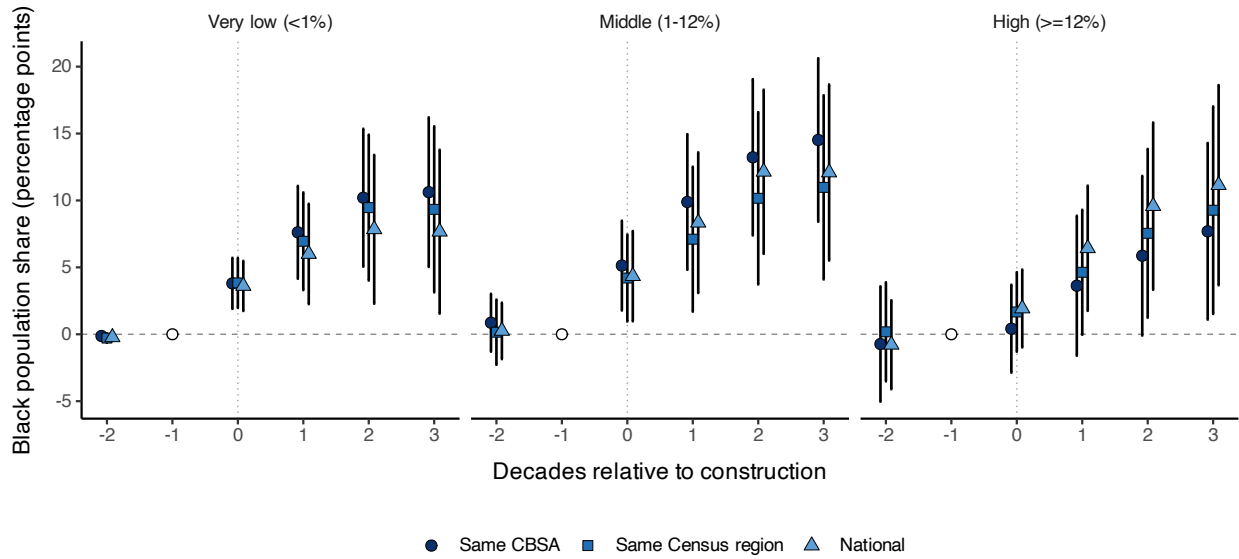
This appendix reports supplementary results and sensitivity checks for the neighborhood analyses.

Figure A2: Black Population Share: Pooled Samples and Weighted Controls Over Event Time



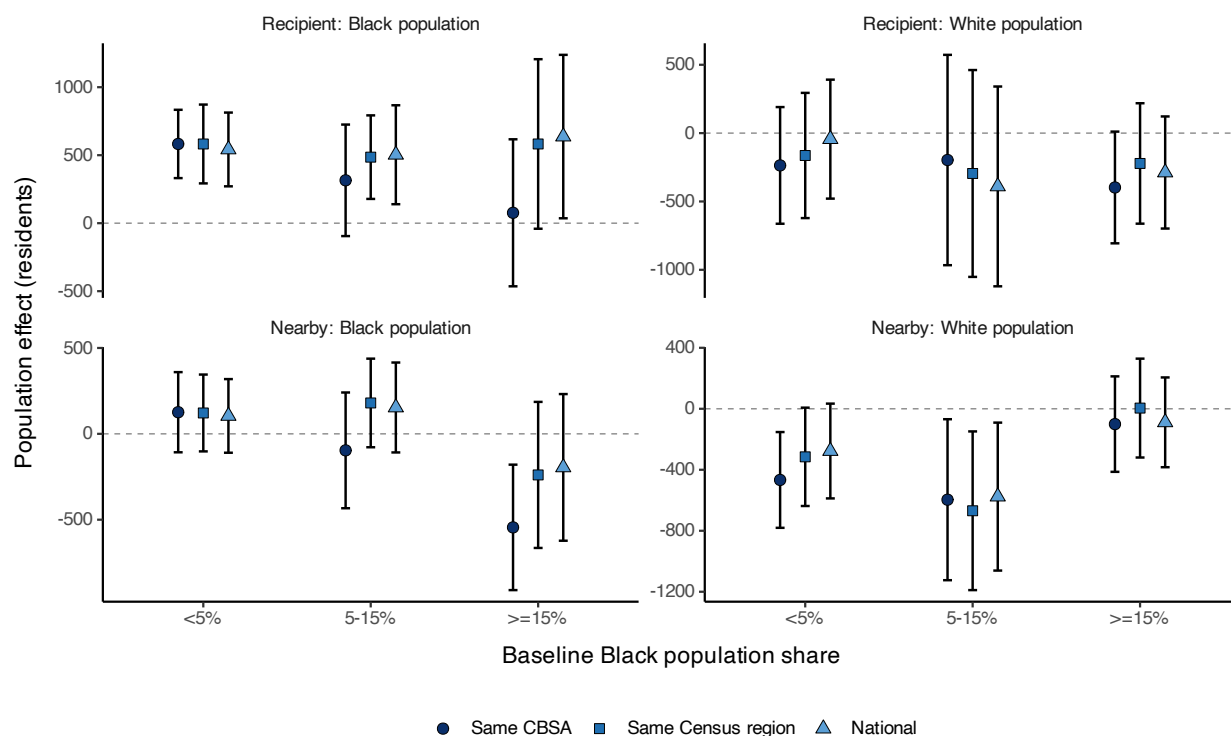
Note: Mean Black population shares for recipient neighborhoods (top row) and nearby neighborhoods (bottom row), with their pooled controls. Controls are matched directly to each recipient or nearby tract from eligible donors in its metropolitan area (same-CBSA), its Census region (same-region), or any metropolitan area in the sample (national). Means use the matching weights, held fixed across event time; lines connect descriptive means rather than event-study coefficients. Each panel uses the corresponding pooled sample. These paths do not establish the absence of spillovers to controls.

Figure A3: Black Population Share Effects by Initial Racial Composition



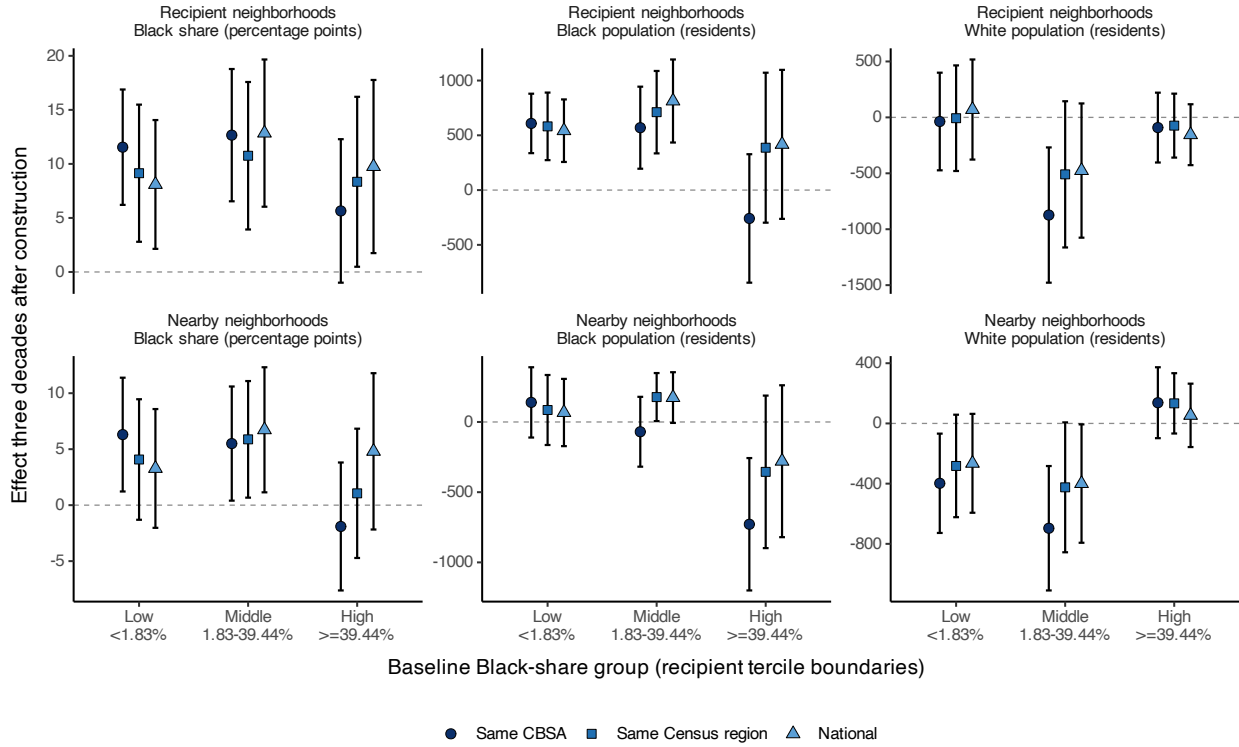
Note: Pooled treatment effects on recipient Black population share, in percentage points, with three selected controls and two pre-treatment decades. Controls are selected from eligible donors in each recipient tract's metropolitan area (same-CBSA), its Census region (same-region), or any metropolitan area in the sample (national). Panels distinguish baseline Black shares below 1% ($N = 299$), 1–12% ($N = 210$), and at least 12% ($N = 494$). Counts are the recipient tracts in each group and are common across these pools. Controls are classified by their own baseline share, with matching weights rescaled within group and construction cohort. Bars are 95% Conley intervals (2 km). The hollow point marks the omitted reference decade $t = -1$.

Figure A4: Racial Population Effects Using 5% and 15% Baseline Black-Share Cutoffs



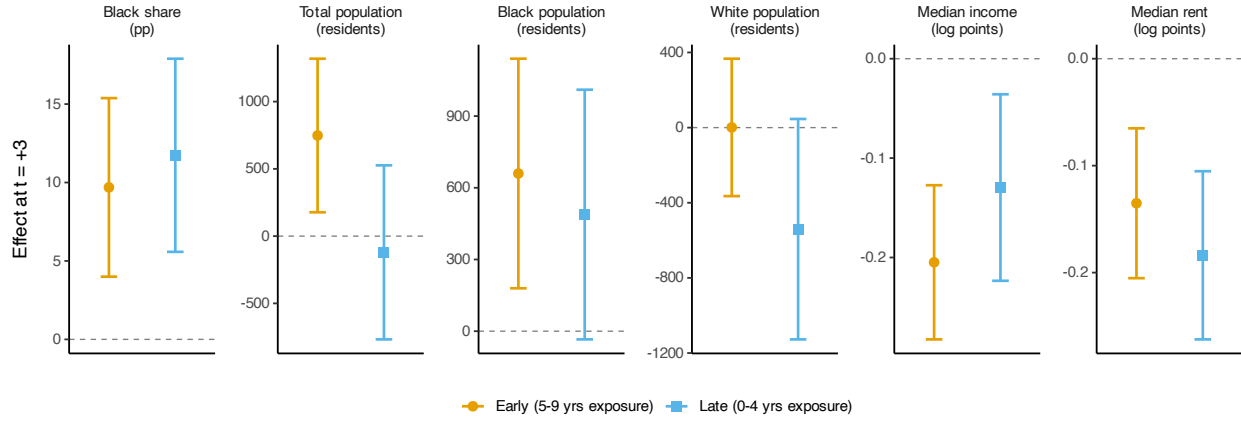
Note: Pooled effects on Black and white population counts, in residents, three decades after construction, for recipient and nearby neighborhoods with three selected controls. Controls are matched directly to each recipient or nearby tract from eligible donors in its metropolitan area (same-CBSA), its Census region (same-region), or any metropolitan area in the sample (national). Categories use baseline Black shares below 5%, 5–15%, and at least 15%. Recipient counts are 416, 122, and 465; nearby counts are 1,949, 288, and 744. The same cutoffs classify donors by their own baseline share. The weighted linear models retain the pooled donor samples and matching weights. Bars are 95% Conley intervals (2 km); the omitted reference decade is $t = -1$.

Figure A5: Racial Population Effects Using Baseline Black-Share Terciles



Note: Estimated effects three decades after construction on Black population share (percentage points) and Black and white population (residents). Top panels show recipient neighborhoods; bottom panels show nearby neighborhoods. Groups use the unweighted terciles of Black population share among 1,003 recipient neighborhoods in the reference decade, with boundaries of 1.83% and 39.44%. The same fixed boundaries classify each nearby neighborhood and donor using its own baseline share. Recipient counts are 334, 334, and 335; nearby counts are 1,637, 846, and 498. Estimates retain the matched samples with three selected controls, two pre-treatment decades, and the relative matching weights. Donors are in the same baseline-share group, with weights rescaled within group and treatment cohort. Models include unit-stack and group-by-treatment-cohort-by-calendar-year fixed effects. Points distinguish eligible donors matched directly to each recipient or nearby tract from its metropolitan area (same-CBSA), its Census region (same-region), or any metropolitan area in the sample (national). Bars are 95% Conley intervals (2 km). The reference decade is $t = -1$.

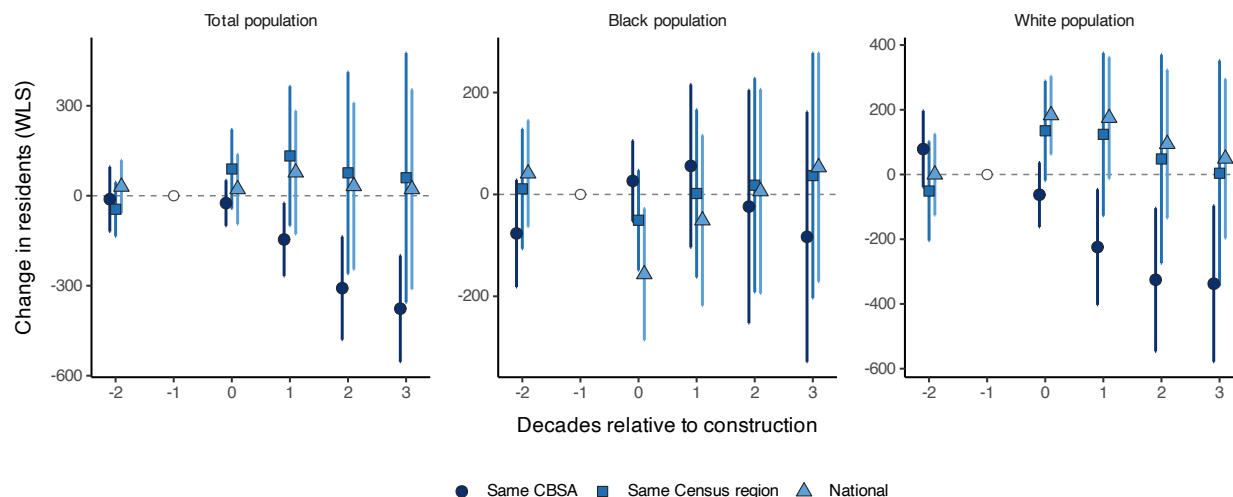
Figure A6: Within-Decade Timing: Early vs Late Openers (Pooled)



Note: Estimated effects three decades after construction ($t = 3$) from Equation 2, with three controls per recipient tract selected from eligible donor tracts across the metropolitan areas in the sample (the national donor pool). Estimates are reported separately for early openers (projects opening 5 to 9 years before the next census, 522 recipient tracts) and late openers (0 to 4 years, 338 recipient tracts). The split pools the 1950, 1960, and 1970 cohorts (860 of 1,003 recipient tracts). The 1980 cohort (143 tracts) is excluded because the project inventory ends in 1973, so every project in that cohort opened 7 to 9 years before the 1980 census and the cohort has no late openers. The six panels show Black population share in percentage points; total, Black, and white population in numbers of residents; and median income and rent in natural logs. 95% confidence intervals based on Conley standard errors (2km cutoff). Reference period: $t = -1$.

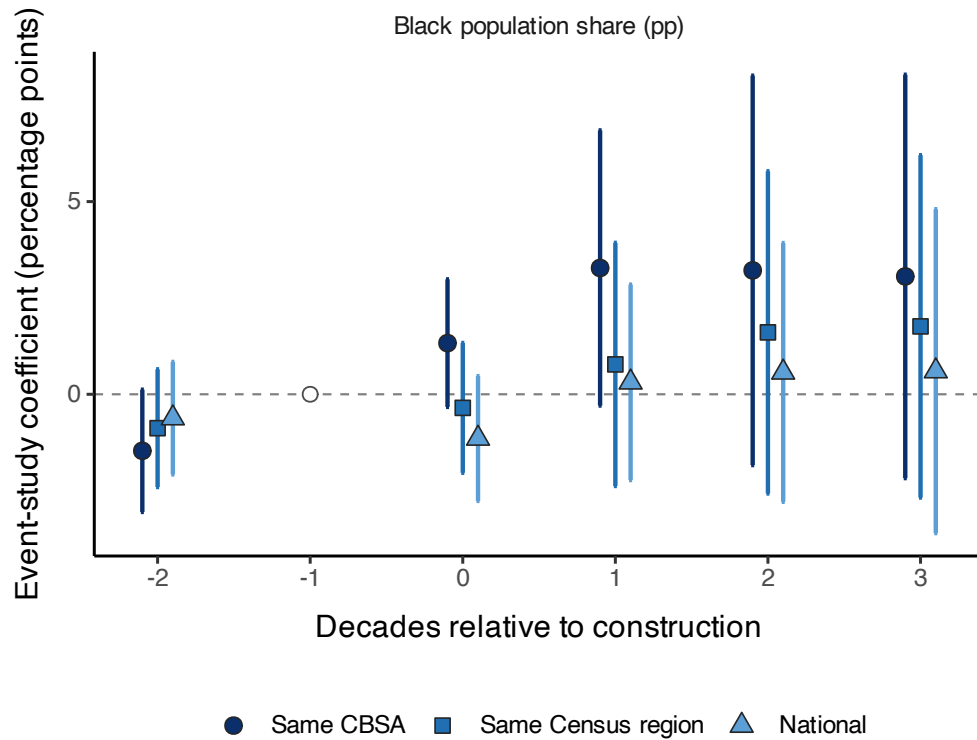
C.1 Placebo-Site Design for Nearby Neighborhoods

Figure A7: Placebo-Site Design: Population by Race in Nearby Neighborhoods



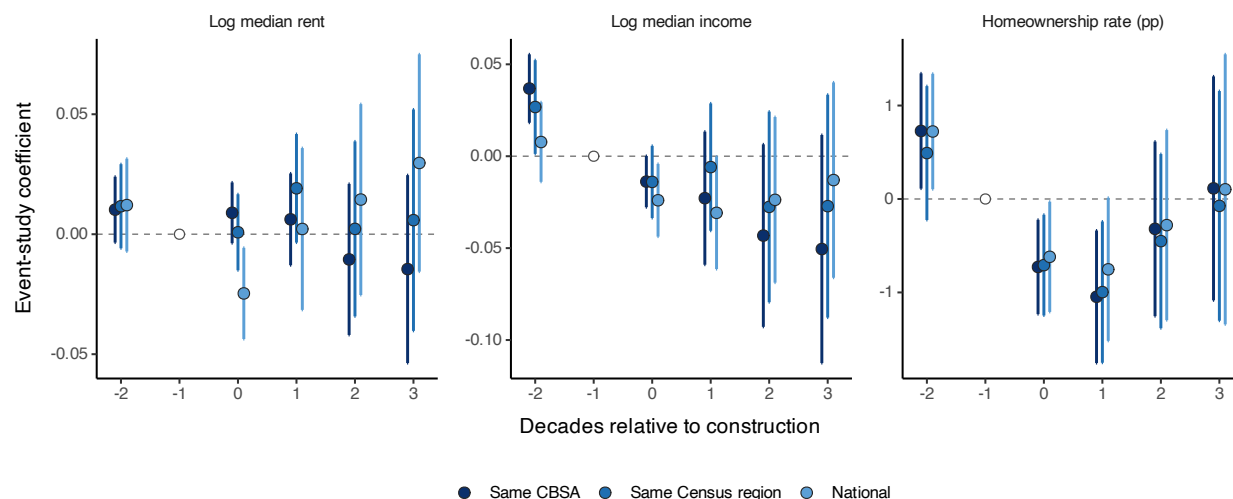
Note: Event-study coefficients from the event-time specification of Equation 2 for nearby tracts under the placebo-site design described in Section 6.1, with tract-by-comparison-set and comparison-set-by-year fixed effects. Three placebo sites are matched to each recipient tract from eligible donors in its metropolitan area (same-CBSA), its Census region (same-region), or any metropolitan area in the sample (national). A comparison set combines the recipient tract's nearby tracts with eligible control tracts within 1 km of the centroids of its placebo sites. Controls exclude recipient and nearby tracts. Total, Black, and white population are measured in numbers of residents. Bars are 95 percent confidence intervals with Conley standard errors (2 km cutoff); the hollow point marks the omitted reference decade $t = -1$.

Figure A8: Placebo-Site Design: Black Population Share in Nearby Neighborhoods



Note: Event-study coefficients from the event-time specification of Equation 2 for nearby tracts under the placebo-site design described in Section 6.1, with tract-by-comparison-set and comparison-set-by-year fixed effects. Three placebo sites are matched to each recipient tract from eligible donors in its metropolitan area (same-CBSA), its Census region (same-region), or any metropolitan area in the sample (national). A comparison set combines the recipient tract's nearby tracts with eligible control tracts within 1 km of the centroids of its placebo sites. Controls exclude recipient and nearby tracts. The outcome is Black population share in percentage points. Bars are 95 percent confidence intervals with Conley standard errors (2 km cutoff); the hollow point marks the omitted reference decade $t = -1$.

Figure A9: Placebo-Site Design: Economic and Housing Outcomes in Nearby Neighborhoods



Note: Event-study coefficients from the event-time specification of Equation 2 for nearby tracts under the placebo-site design described in Section 6.1, with tract-by-comparison-set and comparison-set-by-year fixed effects. Three placebo sites are matched to each recipient tract from eligible donors in its metropolitan area (same-CBSA), its Census region (same-region), or any metropolitan area in the sample (national). A comparison set combines the recipient tract's nearby tracts with eligible control tracts within 1 km of the centroids of its placebo sites. Controls exclude recipient and nearby tracts. Outcomes are log median rent, log median income, and the homeownership rate in percentage points. Bars are 95 percent confidence intervals with Conley standard errors (2 km cutoff); the hollow point marks the omitted reference decade $t = -1$.

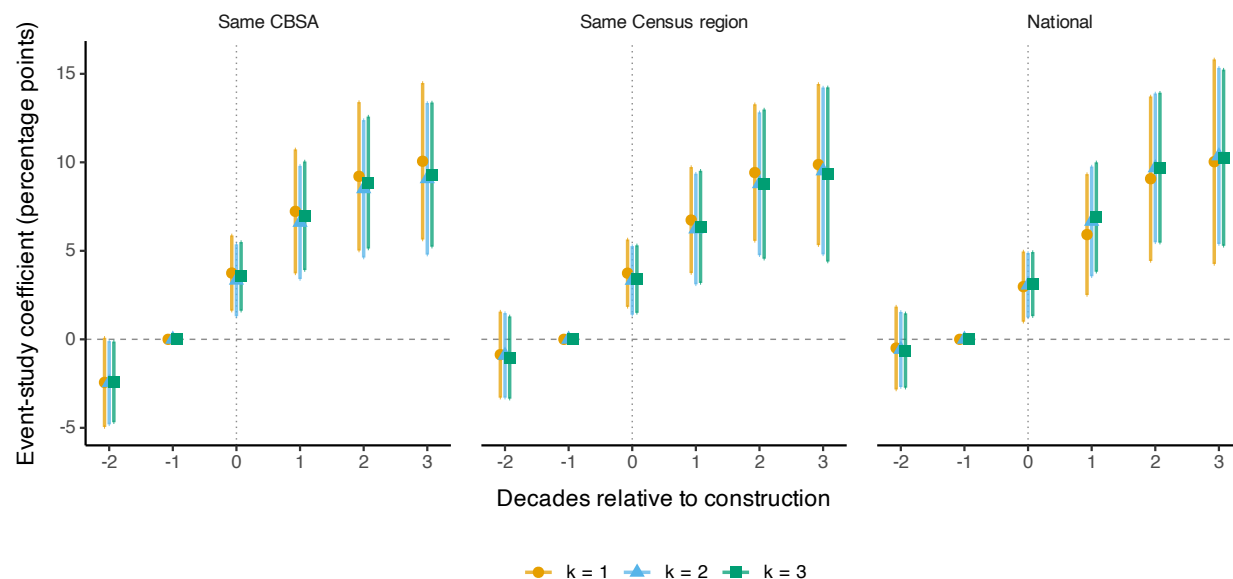
Table A3: Covariate Balance: Placebo-Based Nearby Controls

Characteristic	Nearby tracts			Placebo-based controls		
	CBSA	Region	National	CBSA (SMD)	Region (SMD)	National (SMD)
<i>Matching covariates</i>						
Total population (log)	8.16	8.16	8.16	8.17 (0.011)	8.23 (0.084)	8.27 (0.145)
Black population share	16.1	16.1	16.2	14.9 (0.041)	16.1 (0.001)	18.1 (0.069)
Black population (asinh)	4.50	4.51	4.51	4.43 (0.023)	4.68 (0.058)	4.96 (0.154)
Labor-force participation rate	58.3	58.3	58.4	58.3 (0.009)	58.2 (0.020)	57.9 (0.071)
Unemployment rate	11.5	11.5	11.5	10.9 (0.067)	11.4 (0.010)	10.5 (0.106)
Median family income (log)	3.17	3.17	3.17	3.21 (0.110)	3.17 (0.007)	3.13 (0.084)
Median contract rent (log)	5.79	5.79	5.79	5.83 (0.107)	5.78 (0.021)	5.74 (0.130)
Homeownership rate	31.0	31.1	31.1	34.6 (0.155)	33.0 (0.087)	33.9 (0.129)
Share under age 18	26.6	26.6	26.6	26.7 (0.010)	26.7 (0.007)	26.8 (0.028)
Distance from CBD (asinh)	7.33	7.31	7.31	7.27 (0.016)	6.96 (0.100)	6.84 (0.134)
Redlined (HOLC grade D)	21.4	21.5	21.6	20.6 (0.021)	19.9 (0.040)	21.0 (0.014)
<i>Additional characteristics</i>						
White population (asinh)	8.45	8.45	8.45	8.45 (0.001)	8.57 (0.108)	8.56 (0.106)
Housing needing major repair, 1940	8.8	8.9	8.9	8.6 (0.019)	9.3 (0.042)	9.9 (0.093)
High-school graduate share	30.5	30.5	30.5	32.0 (0.089)	31.7 (0.066)	31.3 (0.048)
Distance to highway (km)	1.74	1.74	1.74	1.74 (0.005)	1.69 (0.032)	1.61 (0.082)
Urban renewal area	16.3	16.4	16.4	8.4 (0.243)	9.5 (0.204)	10.1 (0.187)
Tract-cohort observations	2,920	2,940	2,937	4,120	5,450	5,805
With 1940 repair data	2,811	2,829	2,826	4,014	5,257	5,598

Notes: Means are shown with absolute standardized mean differences (SMDs) from the corresponding nearby sample in parentheses beneath control means. Nearby means appear on the left and control means on the right; each pool has its own retained samples. A recipient tract is retained only when both its nearby area and the area around its matched controls are nonempty. Headings give the donor pool for three placebo sites per recipient. Controls lie within 1 km of these sites' centroids, excluding recipient and nearby tracts. Matching weights are rescaled by recipient tract; nearby tracts receive unit weight per cohort. Counts are distinct tract-cohort observations. SMDs use each comparison's weighted pooled standard deviation. Census characteristics use the preceding census; repair uses 1940 for every cohort. Shares and rates are percentages; income is in thousands of 2000 dollars, rent in 2000 dollars, and CBD distance in meters before transformation; asinh denotes the inverse hyperbolic sine. Additional characteristics are excluded from matching. Repair measures the share of housing needing major repair; only this row excludes missing observations. Redlined denotes at least 80% HOLC grade-D coverage. Highways use the 1996 network; urban renewal denotes at least 5% historical project-area overlap and may coincide with or follow treatment.

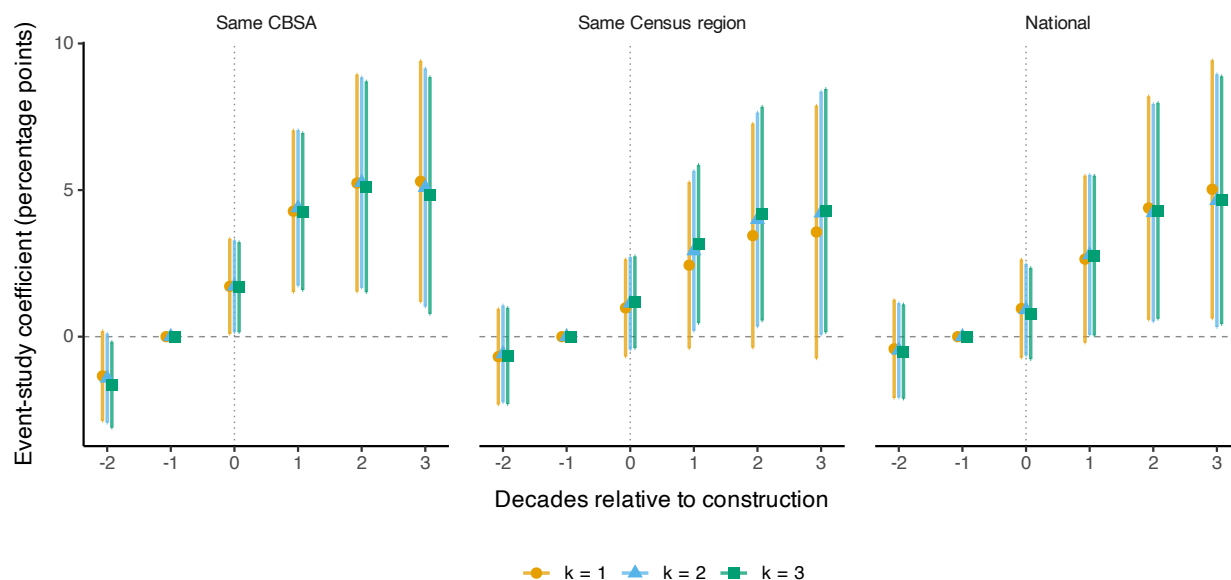
C.2 Sensitivity to the Number of Matched Controls

Figure A10: Black Population Share in Recipient Neighborhoods: $k = 1, 2$, and 3 Matched Controls by Donor Pool



Note: Event-study coefficients from Equation 2 for the Black population share of recipient tracts, in percentage points, re-estimating the main design with one, two, or three matched controls per recipient tract (Section 6.1.1). Panels distinguish controls selected from eligible donors in each recipient tract's metropolitan area (same-CBSA), its Census region (same-region), or any metropolitan area in the sample (national). Bars are 95% confidence intervals with Conley standard errors (2 km cutoff); the reference decade is $t = -1$.

Figure A11: Black Population Share in Nearby Neighborhoods: $k = 1, 2$, and 3 Matched Controls by Donor Pool



Note: Event-study coefficients from Equation 2 for the Black population share of nearby tracts, in percentage points, re-estimating the directly matched nearby design with one, two, or three matched controls per nearby tract (Section 6.1). Panels distinguish controls selected from eligible donors in each nearby tract's metropolitan area (same-CBSA), its Census region (same-region), or any metropolitan area in the sample (national). Bars are 95% confidence intervals with Conley standard errors (2 km cutoff); the reference decade is $t = -1$.

C.3 Spatial Inference

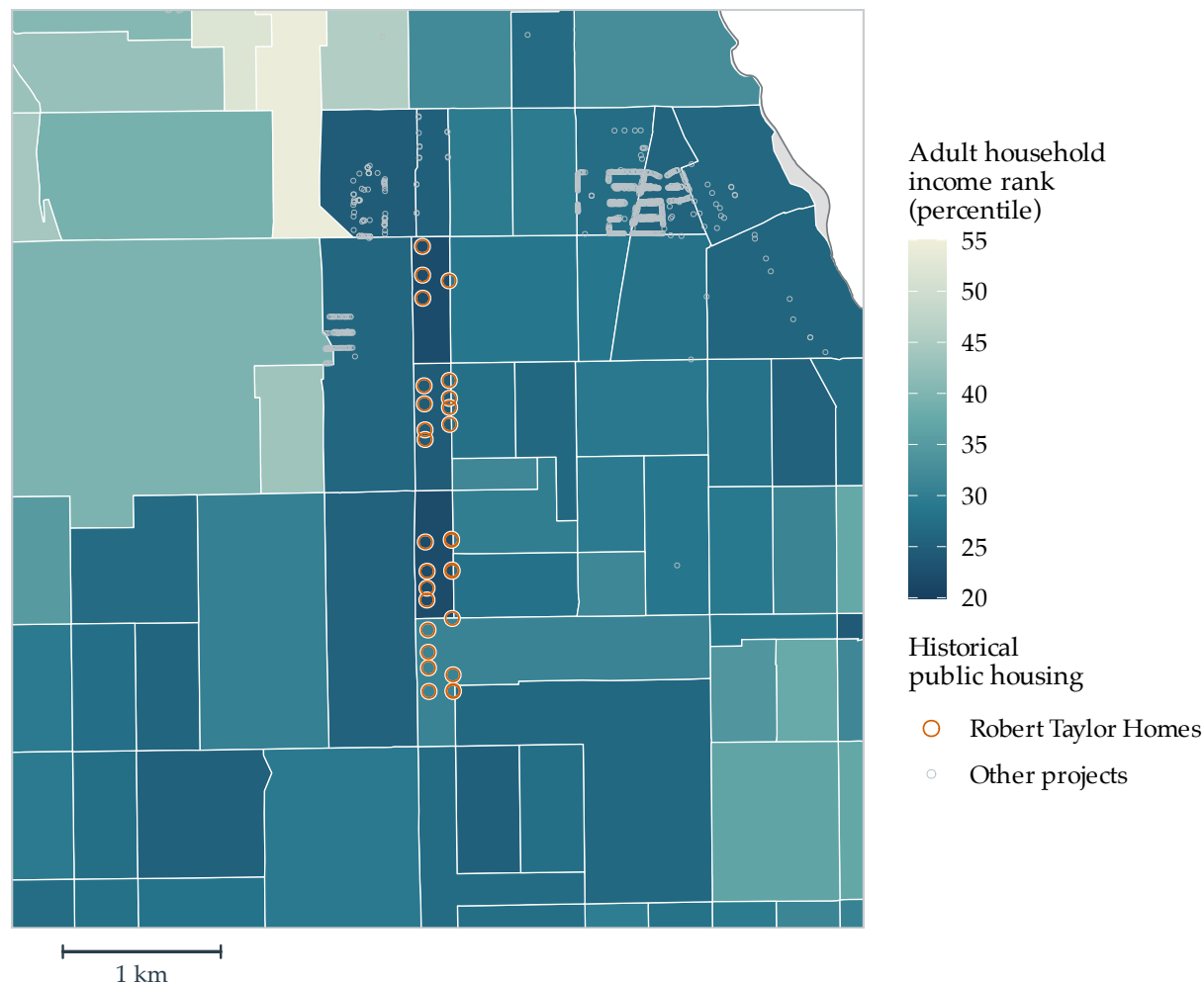
Table A4: Sensitivity to the Conley Distance Cutoff

Outcome	CBSA			Region			National		
	Est.	Conley SE		Est.	Conley SE		Est.	Conley SE	
		2 km	5 km		2 km	5 km		2 km	5 km
<i>A. Recipient neighborhoods</i>									
Total population (PPML)	0.016	0.048	0.049	0.107	0.061*	0.076	0.114	0.059*	0.073
Black population (PPML)	-0.046	0.121	0.109	0.289	0.141**	0.160*	0.351	0.133***	0.145**
White population (PPML)	-0.189	0.075**	0.077**	-0.087	0.088	0.099	-0.068	0.082	0.095
Black population share (pp)	9.31	2.08***	2.01***	9.32	2.51***	2.92***	10.26	2.54***	3.14***
Median income (log)	-0.228	0.044***	0.036***	-0.176	0.036***	0.029***	-0.161	0.034***	0.031***
Median rent (log)	-0.214	0.038***	0.035***	-0.194	0.036***	0.041***	-0.158	0.033***	0.038***
Homeownership rate (pp)	-5.03	1.03***	1.00***	-5.76	0.91***	0.94***	-5.94	0.89***	0.96***
<i>B. Nearby neighborhoods</i>									
Total population (PPML)	-0.132	0.034***	0.029***	-0.072	0.041*	0.046	-0.096	0.041**	0.050*
Black population (PPML)	-0.328	0.135**	0.125***	-0.075	0.141	0.159	-0.014	0.139	0.156
White population (PPML)	-0.223	0.060***	0.055***	-0.131	0.064**	0.064**	-0.120	0.061**	0.063*
Black population share (pp)	4.82	2.06**	1.91**	4.30	2.12**	2.39*	4.66	2.16**	2.61*
Median income (log)	-0.064	0.038*	0.040	-0.024	0.033	0.035	-0.009	0.030	0.034
Median rent (log)	-0.030	0.026	0.023	0.004	0.024	0.027	0.005	0.023	0.026
Homeownership rate (pp)	-0.18	0.68	0.67	-0.50	0.62	0.62	0.07	0.62	0.69

Notes: Effects three decades after construction ($t = 3$), relative to $t = -1$, using three matched controls and the indicated donor pool. Each estimate has Conley standard errors (SE) at 2 km and 5 km; samples, matching weights, and fixed effects are identical. Population rows report PPML log-link coefficients; income and rent are in logs, and shares are in percentage points (pp). Nearby neighborhoods use directly matched controls. Stars beside each SE test whether the coefficient differs from zero using that cutoff: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.4 Neighborhood Opportunity

Figure A12: Upward Mobility around Robert Taylor Homes



Note: The map shows upward mobility in 2010 census tracts around Robert Taylor Homes in Chicago. Shading gives expected adult household income rank in the national distribution for children born in 1978–1983 whose parents were at the 25th income percentile, pooling races and genders. Income is measured in 2014–2015. Darker tracts have lower mobility; the color scale spans 20–55. Orange circles mark the 28 Robert Taylor building/address locations used to assign treatment. Gray circles mark other projects, using building/address locations where available and project coordinates otherwise. Markers have fixed sizes and do not represent building footprints. The map describes neighborhood outcomes rather than effects of public housing. Sources: Opportunity Atlas ([Chetty, Friedman, et al. 2018](#)), NHGIS tract boundaries, and the historical housing data described in Section 4.